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Making and breaking of Zn–C bonds in the cases of allyl and propargyl organozincs

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ABSTRACT

This review aims to sum up recent developments in the synthesis of homoallylic and homopropargylic alcohols or amines by using methods based on catalytic or stoichiometric use of metallic zinc for allylation and propargylations of carbonyls or imines. These methods encompass various transformations which are now commonly expressed as Barbier-Grignard type reactions including several chemical or electrochemical processes.

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1. Introduction

Carbon–Carbon bond formation is the fundamental tool of organic synthesis to construct the backbone of organic compounds.

Organozinc reagents are a high-priority crucial part of the synthetic chemist's society. In modern organic synthesis, more or less every natural product synthesis involves at least one step where this class of compounds has been utilized. As a result, there must be regular need for the development of novel methods which is simple and efficient for the preparation of organozinc reagents. Among various organometallics, organozinc reagents have been established as an

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outstanding reagent which is generally used for the cross-coupling reactions for their functional group tolerance. Considerably, most of the organozinc reagents are simply prepared either by the direct insertion of metallic zinc or transmetalation by zinc salts [1].

Zinc is a very cheap and relatively less toxic metal as well as it is an essential element for humans and every type of animals and plants. In spite of being recognized for over 150 years [2], in organic synthesis, organozinc reagents were applied in limited to extremely rare cases because of their moderate reactivity. Organomagnesium or organolithium reagents are more reactive and in organic synthesis they were found to have broad applications. However, it became obvious that this high reactivity also has a number of drawbacks, like lower chemoselectivity and extremely high reactivity in case of several functional groups. Moreover, the low reactivity of organozinc reagents could be increased by adding transition metal catalyst [3]. Recently, organozinc reagents have been emerged as potent applicant in the field of organic synthesis [4] as well as they are very important reagents in organic chemistry [5]. Even though almost ignored for over hundred years since their discovery, nowadays organozinc reagents have been growing in scope and in importance and become one of the most valuable and useful class of organometallic reagents for their easy preparation and capability of higher functional group tolerance.

In compare to the polarity of C–Li and C–MgX bonds, the C–Zn bond is extremely covalent and thus less reactive, allowing the synthesis of several functionalized derivatives. The C–Zn bond is polarized towards carbon because of the electronegativity differences between carbon (2.55) and zinc (1.65). The application of organozinc reagents is primarily focused on the preparation and use of functional organozinc compounds in organic synthesis (Reformatsky reactions), cyclopropanations (Simmons-Smith reactions), and transmetalation through transition metals.

In 1849, the synthesis of diethylzinc by Frankland [2,6], and the worth mentioning work of Grignard on organomagnesium reagents [7] have paved the mode for the development of fundamental modern organometallic chemistry. In this time, organometallic reagents became the most dominant accessible tool to organic chemists. Depending on the extreme nature of the metal, reactivity and selectivity can be tuned. Although, organolithium species is highly reactive, hardly compatible with sensitive functional groups and shows a poor selectivity, the utilization of less reactive reagents such as organozinc, -tin, or -boron reagents greatly increases due to their tolerance towards functionalities.

Diethylzinc (Et_2Zn) was the first organozinc ever prepared, as well as the earliest compound having metal-carbon sigma bond. A lot of organozinc species are pyrophoric in nature and hence very difficult to handle. Generally, Organozinc compounds are sensitive to oxidation, dissolve in extensively various solvents but often decomposed in protic solvents. In several reactions these reagents are prepared *in situ*.

Although the application of allyl organometallic reagents designed in stereoselective synthesis for the construction of C–C bond have achieved considerable attention over the past decades [8], various issues related with the generation of allyl organometallic compounds still remain unsolved and searching of novel methods continues to be a matter of investigation. For the preparation of masked allylzinc species, reaction of a sterically hindered homoallyl alcohol with *n*-BuLi and after that with ZnCl_2 provided the zinc alkoxide which fragmented *in situ* to generate an allylic zinc reagent (Scheme 1) [9].

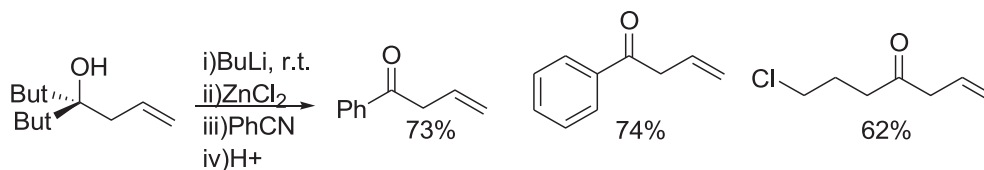
Metal-mediated allylation and propargylation are well-known examples of organic reactions in aqueous media [10]. But excessive metal is used in most of the cases and the corresponding metal salt is generated as waste material. Lots of research has been done on developing metal mediated allylation and propargylation

reactions in aqueous media. Zinc is very cheap and low toxic among the successful metals to mediate this Barbier-type reactions [11]. On the preparation and reactivity of organozinc species [12], it is observed that under mild conditions, various organozinc reagents undergo the cross-coupling reactions with the appropriate coupling partners to afford the building block compounds. Within the earlier two decades, extensive research has been done to develop carbon-carbon bond formation reactions such as allylation and propargylation reactions with organometallic reagents in aqueous media and various metals have been demonstrated to efficiently mediate these reactions. Among them, zinc, a relatively cheap and less toxic metal, has been established as an excellent reagent to mediate such Barbier-type reaction and has been utilized in industry production. Here I wanted to focus on allylic and propargylic organozinc species, the two main classes of organozincs.

As homoallyl alcohols are important building blocks and adaptable synthons, they become extremely featured in the field of organic synthesis of several natural products and biologically active molecules [13]. Metal-mediated allylation reaction is one of the easiest and most convenient, among the existing way to synthesize synthetically and biologically significant molecules [14]. From 1980s, it has been progressively realized that allylation reaction of carbonyls through a Barbier-type condition can be achieved even in aqueous media [15]. Up to now several metal mediated effective coupling reactions of carbonyls with allyl halides has been reported to afford the corresponding homoallyl alcohols in aqueous media. Consequently the metallic zinc mediated carbonyl allylation reactions has also been extensively studied [16]. But the overall yield was not so attractive and the substrate scope were very limited. A few reactions were found to proceed with the aid of other catalyst or ultrasonic irradiation and the reaction required lengthy time. As a part of the rapidly emerging field of green chemistry, more benign sustainable synthetic procedures are attractive [17]. Various approaches are included to avoid the utilization of organic solvents as the reaction media to reduce the amount of waste, energy usage, and the use of volatile, toxic and flammable solvents [17]. Replacement of reaction media incorporate non volatile and recyclable ionic liquids [18], water [19], supercritical carbon dioxide [20], polyethylene and polypropylene glycol [21]. Another alternative approach includes the utilization of the so-called solvent free synthesis [22,23].

Homopropargylic alcohols also have received significant interest as a potent applicant in organic synthesis and synthetic intermediates and as the structural moiety (building block) in various biologically active molecules [24]. The existence of alkyne unit makes it convenient for valuable organic synthesis. A variety of metals have been utilized to generate propargylic organometallics which is reacted with carbonyls for the preparation of homopropargylic alcohols [25–33]. Though, such reactions involved the utilization of water, either pure or with co-solvent, as a medium for reactions with organic or organometallic reagents [34]. Allenylzinc reagents are usually prepared by the oxidative addition by metallic zinc or transmetalations [35–37]. Metallic Zn mediated propargylation reactions with allenyl zinc reagents have received important synthetic utility [36] because of the high stereo, regio, and chemoselectivities of organozinc chemistry [35,38].

Various methods of propargylation of carbonyls with propargylic bromide, such as utilizing effective catalysts, aqueous phase reactions [39], microwave [40], and ultrasound irradiation [41], have been reported. The Barbier-type reactions have been extensively used for the preparation of homopropargyl alcohol in laboratories and industries. The metal or metal compounds mediated Barbier type nucleophilic addition to carbonyls of functionalized halides is treated as an important approach for C–C bond formation



Scheme 1. Preparation and use of masked allylzinc species

in organic synthesis. Considerable attention has been focused on the synthesis of homopropargylic alcohols through Barbier type addition of simple functionalized propargylic bromide to aldehydes or ketones mediated by different metals and their compounds. As a result, reports are available regarding asymmetric [42] as well as non-asymmetric propargylation reactions of carbonyls directly promoted with metals such as indium [43], zinc [44], low valent barium [45] etc. Moreover, there are some reports for homopropargylic alcohol synthesis in other way viz. by zinc catalyzed addition of propargylic boronates to aldehydes [46], through manganese mediated asymmetric propargylation of aldehydes catalyzed by Cr(salen) complex [47] or titanocene (III) complex [48], addition of 1-substituted propargyl mesylate to carbonyls in the presence of $\text{SnI}_2/\text{TBAI}/\text{NaI}$ etc. [49] Additionally, a report is also available for the electrochemical propargylation reaction of carbonyls by using platinum cathode and zinc or aluminum as anode [50]. From the literature it has been established that, several metal mediated propargylation reactions of aldehydes were associated with the formation of the desired homopropargyl alcohols along with the undesired allenyl alcohols to various extents. This is due to the metallotropic rearrangement reactions between propargylic metal and the corresponding allenyl metal during reaction period. In this regard, the regioselectivity in the synthesis of homopropargylic alcohol has been improved during propargyl-Grignard reaction of carbonyls by adding poisonous HgCl_2 catalyst since early days or performing the reaction with ZnBr_2 following a careful reaction protocol [51]. In another remarkable approach, homopropargyl alcohol could be synthesised with high regio-selectivity during the reaction of carbonyls with allenylmetals [42,52]. Needless to say, the synthesis of homopropargylic alcohol by metal mediated Barbier type addition of propargylic bromide to carbonyls is always attractive from a practical point of view. Still there remain further scope to explore the different metals in their different forms and their ability to promote direct addition of propargylic bromide to carbonyls. The regiocontrolled synthesis of homopropargyl alcohols has been extensively studied; there are still not many satisfactory methods for the selective propargylation [53].

2. Allylic zinc reagents

2.1. Introduction

Compared to Grignard and organolithium reagents, organozinc reagents are easy to prepare and handle [54]. Since the discovery of the formation of alkylzinc compounds by Frankland from the reaction of iodoalkanes with zinc metal [2], the application of organometallic reagents in organic synthesis has been rising in scope and in significance [55]. The achievements of Reformatsky [56], Barbier [57], Grignard [58], and Gilman [59] are the milestones in the progress of this field. Further recent development includes the utilization of Li and other metals. In their most general form, these reagents are being used as nucleophiles, which lead to C–C bond formations. When carbonyl compound is the reactant electrophile, the transformations are now commonly expressed as Barbier-Grignard type reactions. The organometallic reagents can

be generated *in situ* (Barbier-type) or stepwise (Grignard-type). It was supposed that strict exclusion of moisture is the mandatory requirement in such cases. Such a restriction can impose limitations on synthetic design in which a variety of acidic hydrogens in the substrates have to be protected.

In 1977, Wolinsky and co workers reported that zinc mediated carbonyl allylation reactions with allyl bromide could be performed in 95% ethanol and *t*-butyl alcohol [60]. But, they got just a moderate yield. Since that time, considerable development has been done which shows that the reaction can be carried out in aqueous medium by the use of various metal mediators. These metals include Zn, Sn, In, Bi, Pb, and Cd. It can be updated that these metals are relatively “soft” and are not readily affected by the “hard” solvent water.

2.2. Generation and reactivity of allylzinc

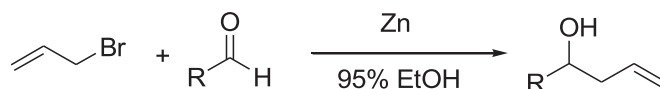
In 1977, Wolinsky and co workers reported a pathway for allylation reaction of *in situ* generated allylzinc bromide with carbonyls. Here allyl bromide was added slowly to stir slurry of “activated” zinc dust and carbonyl in ethanol or *tert*-butyl alcohol at 78–95 °C to afford desired homoallylic alcohols in preparative yields analogous to that achieved in aprotic solvents. Previous to this information, Wurtz-type coupling of allylic halides was (in lower yields) the usual results in refluxing alcohol (Scheme 2) [60,61].

In 1978 Bellassoued et al. reported a valuable synthetic route to prepare allylzinc bromide through the reaction of allyl bromide and metallic zinc in THF for 1 h at ambient temperature (Scheme 3) [62].

After that, in 1985 Luche and co workers, examined that carbonyls can be allylated in water-THF using metallic zinc under ultrasonication [63,64]. The replacement of water enhanced the efficiency. The yield of the reaction increased when water was replaced by aqueous saturated NH_4Cl solution. Here, comparable outcomes were achieved either with or without the utilization of sonication (Scheme 4).

Mattes et al. in the same year, carried out a metallic zinc mediated coupling reaction of ethyl (2-bromomethyl) acrylate with aldehyde or ketones in a mixture of saturated aqueous NH_4Cl -THF under reflux condition to afford the corresponding cyclic lactonization product with 47–98% yield. The reaction yield decreases to 15% when same reaction was performed in THF alone. Even though it becomes much less efficient, (2-bromomethyl) acrylic acid can also be directly added upon neutralization with triethyl amine (Scheme 5) [65].

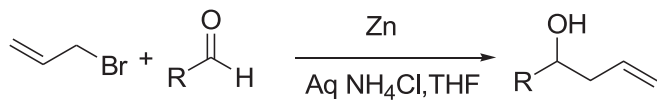
Furthermore, Wilson reported a more detailed study of allylation reactions using allyl halides (bromide or chloride) mediated by zinc in aqueous medium by modifying the use of a solid organic supports included reverse phase C_{18} silica gel, S-X8 bio-beads



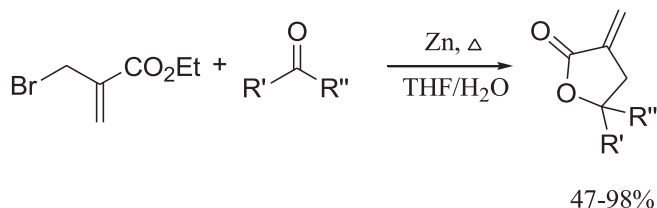
Scheme 2. Allylation reactions of in situ generated allylzinc bromide with carbonyls



Scheme 3. Preparation of allylzinc bromide



Scheme 4. Allylation in water-THF using metallic zinc under ultrasonication



Scheme 5. Zinc mediated allylation and subsequent lactone formation

which is a spherical porous styrene divinylbenzene copolymer with 8% cross-links, and GC column packing OV-101 on Chromosorb, etc. (Scheme 6) [66].

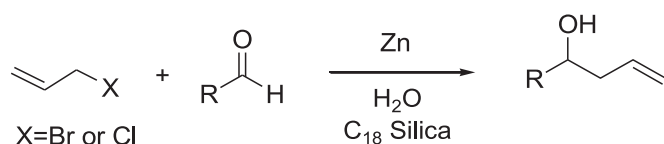
Kunz et al. reported zinc mediated allylation reaction of methyl γ -oxocarboxylates in a mixture of saturated aqueous solution of NH_4Cl and THF. This reaction provides a convenient synthesis of 5-allyl-substituted γ -lactone (Scheme 7) [67].

In 1990, Chan et al. reported that the reaction between carbonyl compounds and 1,3-dichloropropene with zinc in aqueous medium gives conjugated 1,3-butadienes in reasonable yields [68]. Replacement of 3-iodo-1-chloropropene instead of 1,3-dichloropropene enhanced the yields. After the initial allylation reactions, subsequent base treatment of the intermediate compounds produced vinyloxiranes in excellent yields. In the same way, carbonyl compounds reacts with 3-iodo-2-chloromethyl-1-propene followed by base treatment to give 2-methylenetetrahydrofurans [69]. Thus, the 3-iodo-2-chloromethyl-1-propene here served as a novel trimethylenemethane equivalent (Scheme 8) [70].

Oda et al. reported zinc mediated allylation reactions of carbonyls by 2, 3-dichloro-1-propene in a two-phase system of water and toluene containing small amount of acetic acid to afford the corresponding allylation products which were also scalable up to industry label (Scheme 9) [71]. No conversion takes place when tin was used as the promoter. The absence of water completely shuts down the reaction. Interestingly, the treatment of 2,3-dichloropropene with the zinc powder in aqueous ethanol results the dechlorinated allene product.

Very recently, Reisse utilized “activated” zinc achieved by pulsed sono-electro reduction which is about three times more efficient than the commercially available zinc for aqueous Barbier-type reactions [72].

In 2011 Lim et al. examined a zinc-mediated allylation of 2-



Scheme 6. Allylation reactions by the use of a solid organic supports

Methyl-4-oxo-3,4-diphenyl-but-2-enoic acid methyl ester with 5 equivalent zinc dust and 2.5 equivalent allyl bromide in THF at reflux condition (5 h) affording 5-Allyl-3-methyl-4,5-diphenyl-dihydro-furan-2-one in a reasonable yield (70%) (Scheme 10) [73].

In 2006, Bian et al. reported a rapid and high yield allylation of aldehydes and ketones under ultrasonic irradiation [74]. Where aromatic aldehydes or ketones undergo zinc mediated allylation with allyl bromide in the presence of aqueous NH_4Cl –THF under ultrasonic irradiation at room temperature for 10 min to give the corresponding homoallylic alcohols (yields 83–100%) (Scheme 11).

In 2012 Zhang and co-workers reported another novel and special reaction of allylzinc reagent with α -haloketones producing a series of substituted allylic epoxides [75]. Allylic epoxides were synthesized with no additives. After their initial success they proceeded to examine the scope of this reaction. They presented a full account of research that has efficiently prepared a series of epoxides by several allylzinc bromides with α -haloketones (Scheme 12).

Biswas et al. described an important route for the earliest reactions of organozinc reagents with cyclophosphazene derived alkynes which have been performed with allylzinc bromide leading to a variety of novel alkene derived fluorinated cyclophosphazenes. Allylzinc bromide reacts with substituted ethynylferrocene derived fluorophosphazenes to afford the regioselective allylated products (Scheme 13) [54].

In 2015 Lasch et al. gave the first examples of allylation reactions of phenylazocarboxylic esters mediated by Zn [76], that are versatile and extremely rapid to access functionalized hydrazines. Barbier-type allylation reactions of phenylazocarboxylic *tert*-butyl esters have been achieved by using the corresponding allyl and benzyl bromides or iodides in combination with Zn powder in exceptionally short reactions times (Scheme 14). Whereas all reactions take place exclusively at the β -nitrogen atom of the azo-carboxylic esters, the linkage of allyl units was found to depend on the substitution pattern at the double bond of the allyl bromide or iodide. The hydrazines achieved are useful precursors for indoles and indazoles.

Both aldimines and ketimines were found to allylate under Barbier type reaction conditions using allyl bromide and commercially available zinc dust in tetrahydrofuran at room temperature (Scheme 15) [77].

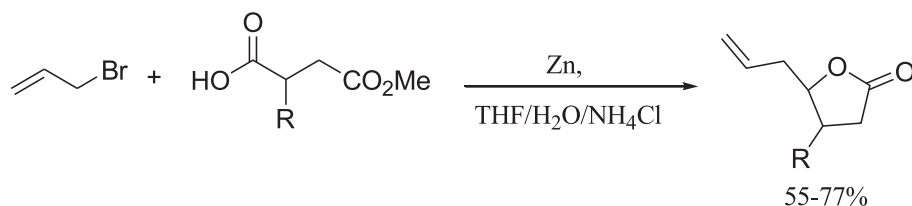
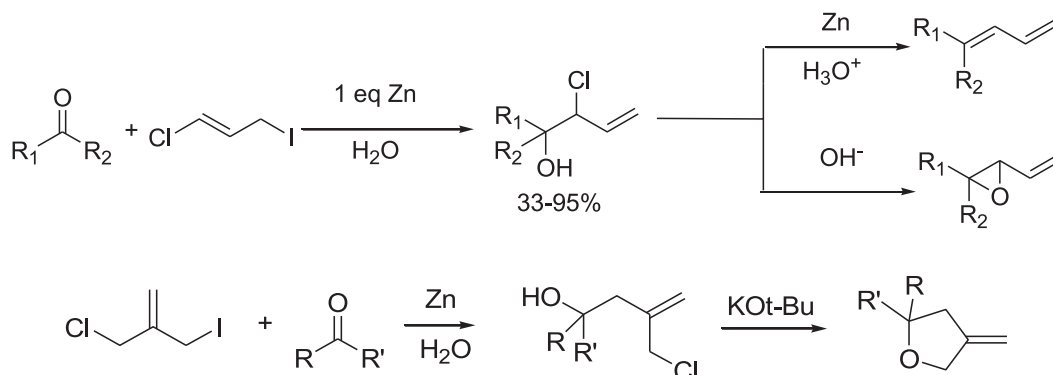
Copper catalyzed tin, aluminum or zinc mediated Barbier-type allylation of iminium ions, generated from secondary amines and formaldehyde in aqueous medium, produces corresponding homoallylamines in up to 85% yield (Scheme 16) [78].

In the year 1985 C. Petrier et al. showed preferential allylation of aldehydes and ketones by the Sn and Zn, simply achieved in aqueous medium (Scheme 17) [79]. Homoallyl alcohols are obtained in good yields by this mechanistically fascinating method.

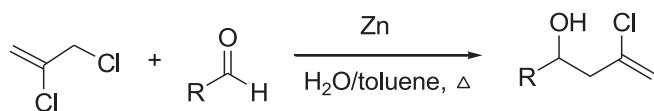
Makosza et al. reported a study of the synthetic utility of the Barbier type allylation of carbonyl compounds with commercial zinc powder in boiling ammonia (-33°C). The reaction of aromatic aldehydes and cinnamic aldehyde with allylic bromides proceeds smoothly to give the corresponding homoallylic alcohols in moderate to high yields. Somewhat lower yields gave the reaction with aliphatic aldehydes and with ketones (Scheme 18) [80].

Breton et al. investigated a zinc-mediated aqueous Barbier–Grignard reaction of cyclic allylic bromide substrates with various aldehydes and ketones to afford homoallylic alcohol with high regioselectivity. Aromatic aldehydes and ketones afforded adducts in good yields (66–90%) and with good diastereoselectivities. Non-aromatic aldehydes also reacted well under these conditions, but only poor yields were obtained with non-aromatic ketones (Scheme 19) [81].

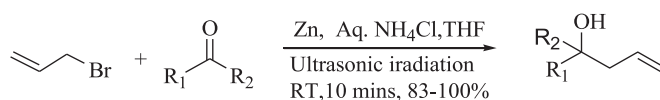
Maeda et al. reported a zinc mediated allylation of aldehydes induced by inorganic and organic salts. Their study indicates that

Scheme 7. Synthesis of 5-allyl-substituted γ -lactone

Scheme 8. Allylation with 1,3-dihalo propene



Scheme 9. Allylation reactions by 2, 3-dichloro-1-propene in a two-phase system



Scheme 11. Allylation of aldehydes and ketones under ultrasonic irradiation

the allylation of aldehydes by zinc powder in MeCN can be effectively performed by the use of Py^+ perchlorate, which appears to activate the zinc surface (Scheme 20) [82].

Rainer Sjöholm et al. suggested in 1994 that a Zn mediated reactions of cinnamyl chloride with aldehydes and ketones in THF- $\text{NH}_4\text{Cl(aq)}$ give besides α - and γ -addition products-phenyl propenes and dicinnamyls. The allylation is initiated by the formation of an allylic radical which then adds to the carbonyl group of the aldehyde (ketone). The addition would result in formation of an alkoxide radical which immediately adds an electron released from the metal and forms the alcohol (Scheme 21) [83].

Zhang et al. reported an efficient, one-pot synthetic approach for the preparation of allylic epoxides, aldehydes and homoallylic alcohols (scheme) The three industrial products were synthesized by the reaction of allylzinc bromide with α -halo ketones under non-catalytic conditions with *p*-toluenesulfonic acid and ytterbium triflate as catalysts, respectively (Scheme 22) [84].

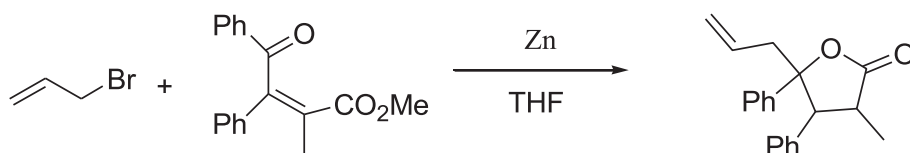
Carbonyl compounds react with organozinc reagents in the presence of diphenyl phosphate to afford the corresponding olefins. A variety of 1,3-dienes and unsaturated esters were obtained in

moderate to excellent yields under mild conditions (Scheme 23) [85].

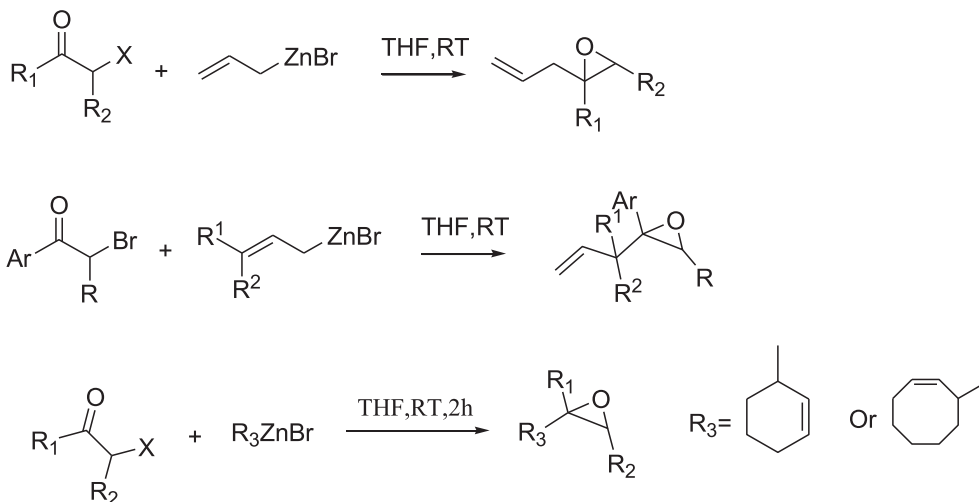
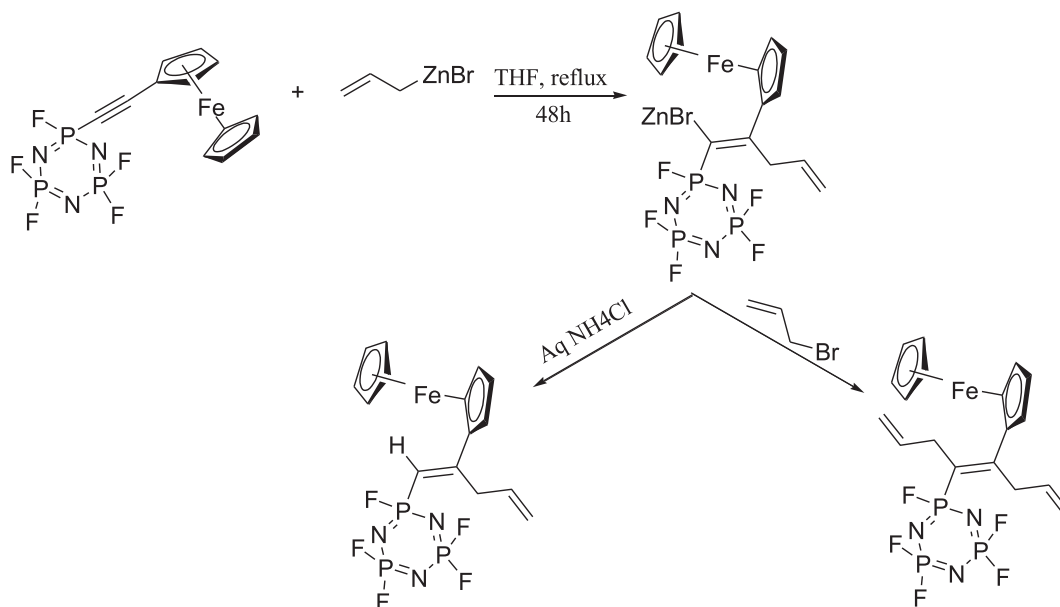
Zhang and co-worker reported a convenient and efficient method for the synthesis of a variety of α -halo, α -allylic aldehydes from α,α -dihalo ketones using both cyclic (3-bromocyclohex-1-ene zinc bromide and (Z)-3-bromocyclobut-1-ene zinc bromide) and acyclic (allylzinc bromide and cinnamylzinc bromide) type of allylic organozinc bromide with DMF as base (Scheme 24) [86].

Zhao et al. reported an efficient and highly α -regioselective strategy for the synthesis of 3-prenyl-3-hydroxy-2-oxindoles has been developed starting from isatins and prenylzinc with good to excellent yields. This protocol provides a straightforward and practical way to introduce an α -prenyl moiety into the C-3 position of isatins. The advantages of this reaction are use of the cheap and readily available reagents, operational simplicity, and wide substrate scope. Furthermore, this transformation was applied to the synthesis of several oxindole-containing natural products, which further demonstrated the synthetic utility of this methodology (Scheme 25) [87].

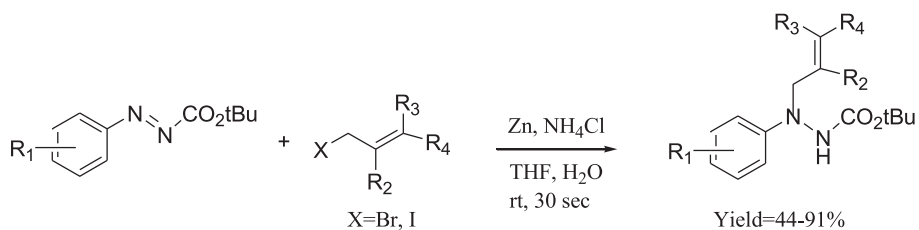
Zhao et al. reported demonstrated a zinc-mediated crotylation of carbonyls to afford α -linear homoallylic alcohols exclusively. In



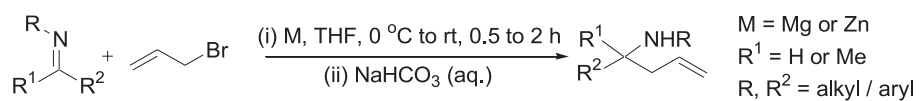
Scheme 10. Zinc-mediated allylation and subsequent lactonisation

Scheme 12. Reactions of allylzinc reagent with α -haloketones

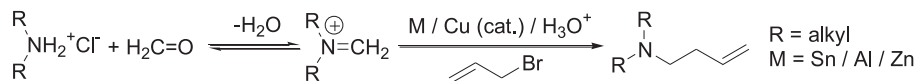
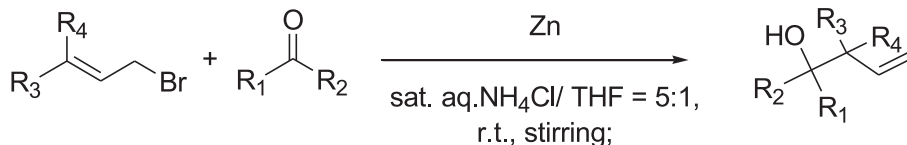
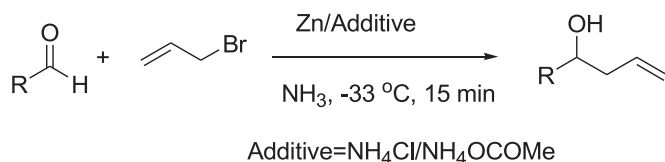
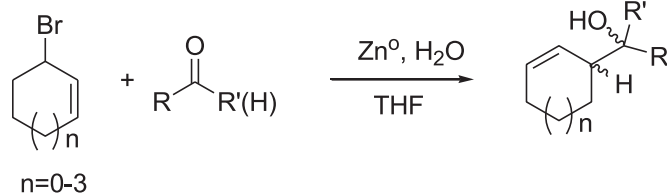
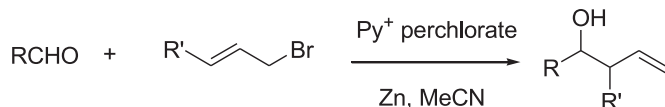
Scheme 13. Reactions of organozinc reagents with cyclophosphazene derived alkynes



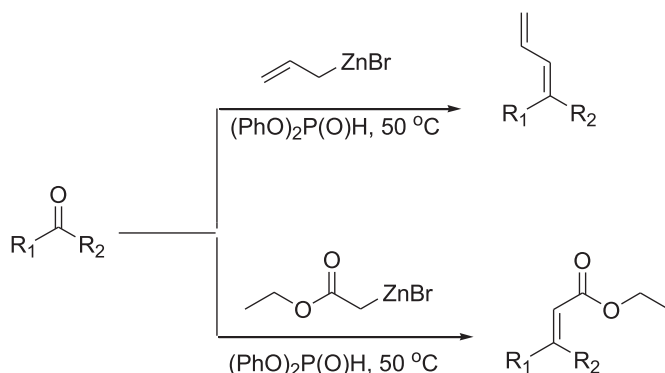
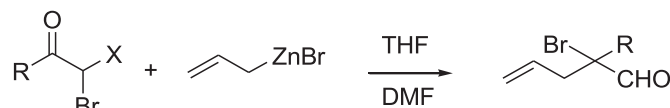
Scheme 14. Allylation reactions of phenylazocarboxylic esters



Scheme 15. Allylation reactions of aldimines and ketimines

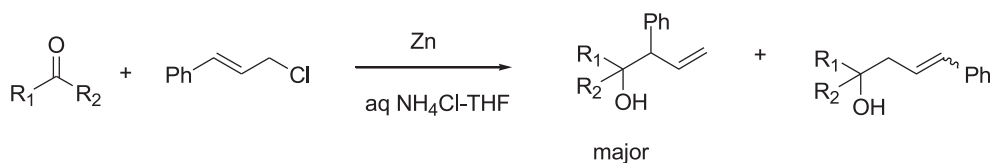
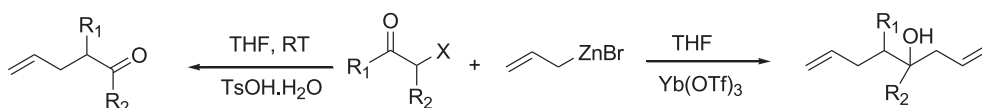
**Scheme 16.** Copper catalyzed tin, aluminum or zinc mediated allylation of iminium ions**Scheme 17.** Preferential allylation of aldehydes and ketones**Scheme 18.** Allylation of carbonyl compounds with commercial zinc powder in boiling ammonia**Scheme 19.** Barbier–Grignard reaction of cyclic allylic bromide substrates**Scheme 20.** Allylation of aldehydes induced by inorganic and organic salts

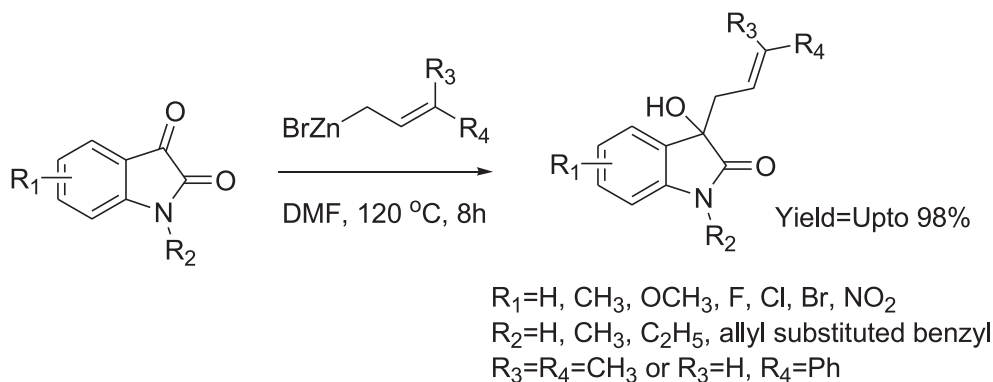
this context, they described a method for the α -regioselective crotylation of aldehydes and ketones by using an easily available starting material, such as crotyl bromide, zinc dust, and the less-toxic solvent DMPU. The reaction has broad substrate scope and is

**Scheme 23.** Reaction of allylzinc bromide in the presence of diphenyl phosphate**Scheme 24.** Synthesis of α -Halo, α -allylic aldehydes by allylic zinc bromide

highly efficient. Moreover, a metallo-[3,5]-sigmatropic rearrangement is proposed to account for the high α -regioselectivity in the zinc-mediated crotylation (**Scheme 26**) [88].

Yin et al. in 2020 reported operationally simple, a ball-milling-enabled zinc-mediated Barbier-type allylation of a broad series of aldehydes and ketones. The reaction is applicable across a range of different zinc metal morphologies (**Scheme 27**) [89].

**Scheme 21.** Allylation by cinnamyl chloride**Scheme 22.** Reaction of allylzinc bromide with α -halo ketones

Scheme 25. Zinc-Mediated C-3 α -Prenylation of Isatins

2.3. Stereoselective allylation

Allyl organometallic species have been studied in detail since the 1960s. In the beginning, efforts were put on the structural determination of allylmetals, like the stereochemistry of the double bond or 1,3-transposition of metals on the allylic system [90]. But after the few years, the discoveries of Gaudemar [91], Heathcock [92], Hoffmann [93], and Yamamoto [94] showed that the stereocontrol of the carbon-carbon bond formation in the reaction of allylmetals with carbonyls could be achieved. Hoffmann noticed that (Z)-crotylboronates lead to *syn*-homoallyl alcohols in a stereoselective approach, and Yoshinori Yamamoto discovered that crotyltins, irrespective of the geometry of the double bond, afford *syn*-homoallyl alcohols under Lewis acid catalysis [93,94].

Keltjens et al. reported the Zn-mediated allylation of 3-formylcephalosporins under aqueous Barbier type conditions using non-activated Zn and saturated aqueous NH_4Cl , in THF at room temperature [95], from which the corresponding homoallyl alcohols were synthesized in good yields and with good to excellent diastereoselectivity (Scheme 28).

The zinc insertion to substituted allylic halides is less effective due to the formation of substantial amount of homocoupling product. Knochel et al. in 2007 described the preparation of substituted allylzinc chlorides (a-e) by the reaction of allyl chlorides with zinc dust in the presence of LiCl in THF with moderate to good yields and their reactivity is also explored (Scheme 29) [96].

The previous methodology can be extended to linear allyl chlorides or phosphates. The study began with the easily available cinnamyl phosphate [97,98], which was treated with 3 equivalent Zn and 3 equivalent LiCl in THF at room temperature for 18 h, complete conversion was obtained and the corresponding allylzinc species was produced in 72% yield without any trace of homocoupling product (Scheme 30). In the absence of LiCl the yield was decreases to 52%, and homocoupling products were isolated in 5–10% yield.

With these organozinc reagents in hand, their reaction with carbonyl derivatives was investigated [99]. When cinnamylzinc phosphate reacted with benzaldehyde in THF at -78°C , the corresponding *anti* homoallyl alcohol was produced within 1 h in good yield and diastereoselectivity (Scheme 31). Amusingly, no

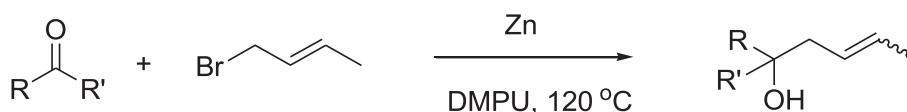
applicable difference in yield or diastereoselectivity was obtained when cinnamylzinc chloride was used instead of phosphate; in this case, and the same product was obtained in similar yield and diastereoselectivity.

Again when the previously prepared allylzinc reagents were treated with substituted acetophenones the corresponding homoallyl alcohols were produced with good yields and diastereoselectivities (yields > 90%; dr > 98: 2) (Scheme 32) [99].

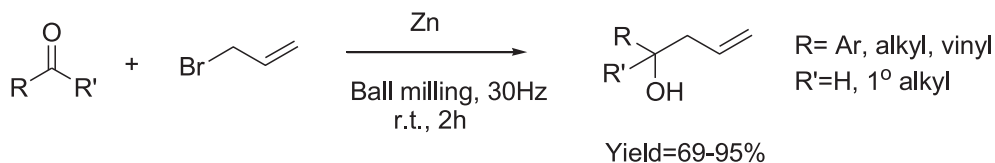
Similarly, when 3-cyclohexyl-prop-2-enylzinc chloride was reacted with 4-bromo-acetophenone, the desired quaternary alcohol was afforded as a single diastereoisomeric product in good yield. The reaction of cinnamylzinc chloride with several methyl alkyl ketones proceeds diastereoselectively as well. Therefore, the addition of 3-methyl-butan-2-one, 1-cyclopropyl-ethanone, 1-cyclohexylethanone or even 1,1,1-trifluoro-propan-2-one to achieve the corresponding alcohols in moderate yields with good diastereoselectivity. Again, when cinnamylzinc chloride was reacted with 1-cyclopropyl-ethanone at 0°C the desired homoallyl alcohol was isolated in good yields with excellent diastereoselectivity. Finally, when 1-trimethylsilyl-ethanone was treated with cinnamylzinc chloride, the corresponding tertiary alcohol was obtained in good yield and diastereoselectivity and no Brook rearrangement [98,100] was observed (Scheme 33).

When the cinnamylzinc reagent reacts with linear methyl alkyl ketones as electrophile, the diastereoselectivity dropped, and single inseparable mixtures of diastereomers could be obtained. Hence, with the electrophile, butan-2-one, corresponding alcohol was isolated in good yield as a mixture of two diastereomers. Remarkably if 2-octanone was used as an alternative, the corresponding alcohol was obtained as a 50: 50 mixture of diastereomers (Scheme 34) [98].

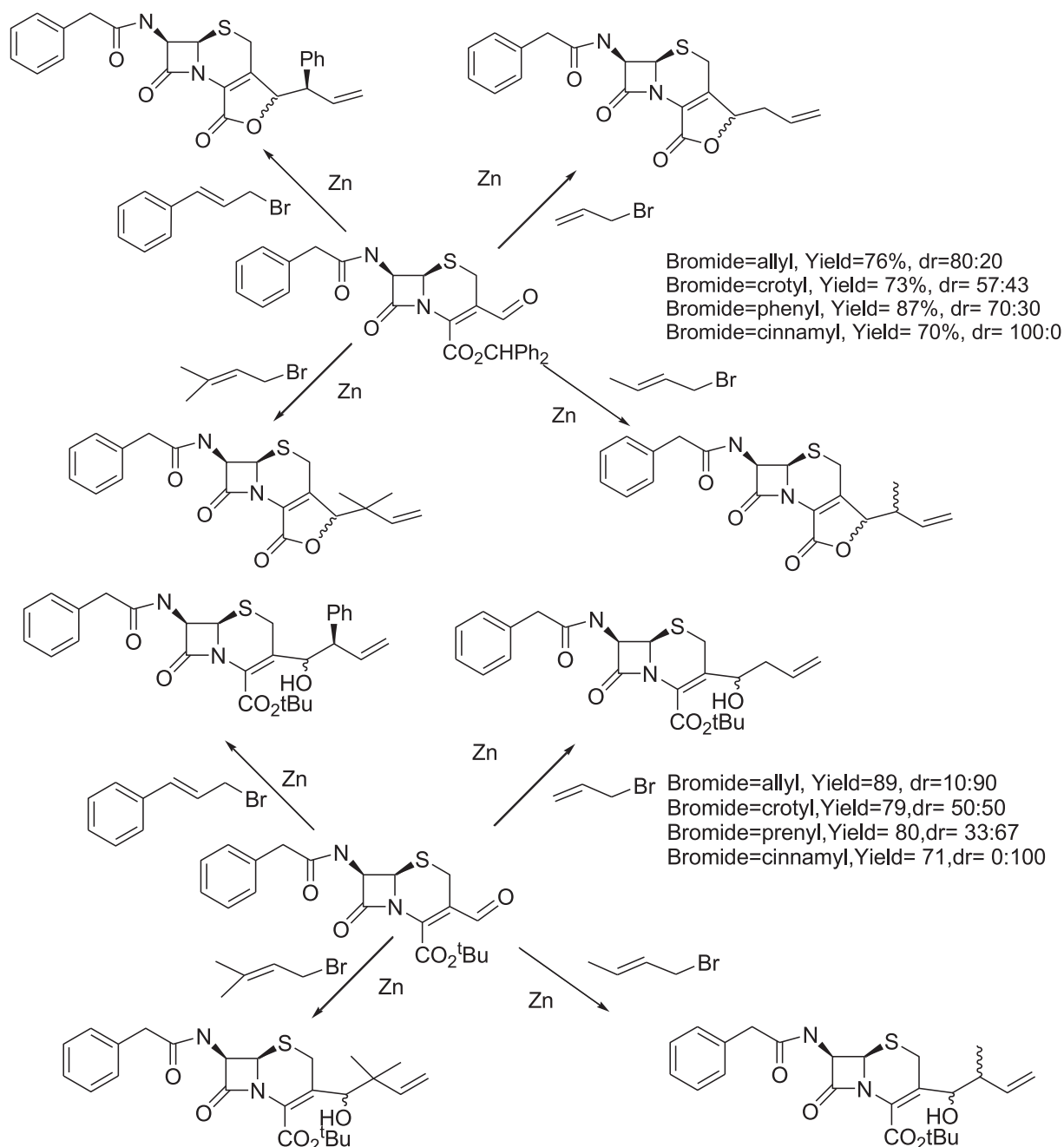
While alkyl vinyl ketones were reacted with the cinnamylzinc reagents in THF at -78°C , the corresponding homoallyl alcohols were obtained with diastereoselectivity depending on the level of substitution of one of the two moieties (Scheme 35). Consequently, when cinnamylzinc reagent reacted with β -ionone, homoallylic alcohol was produced in moderate yield and diastereoselectivity. Amusingly, when 2-iodocyclohex-2-enone was used as electrophile instead, the corresponding homoallylic alcohol was obtained in good yield and diastereoselectivity [98].



Scheme 26. Zinc-mediated crotylation of carbonyls



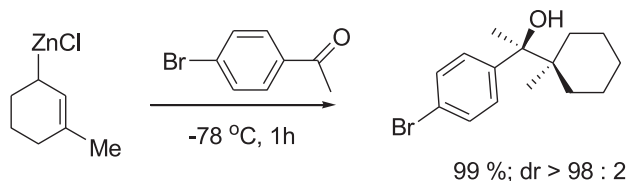
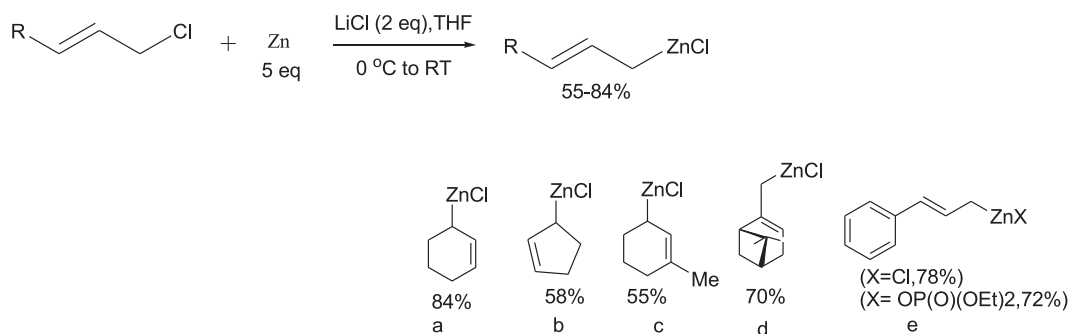
Scheme 27. Ball-milling-enabled zinc-mediated Barbier-type allylation



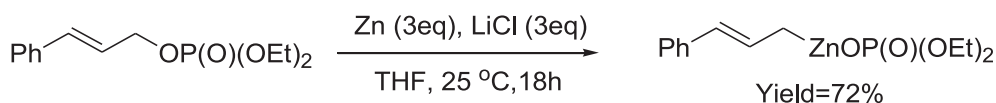
Scheme 28. Zn-mediated allylation of 3-formylcephalosporins

Then the treatment of the bulky trimethylsilyl-substituted allylic zinc reagent with 4-bromo acetophenone under the same reaction conditions is observed in good yield and diastereoselectivity. Unfortunately, in this case, the exact structure of the product could not be assessed with assurance (Scheme 36) [101].

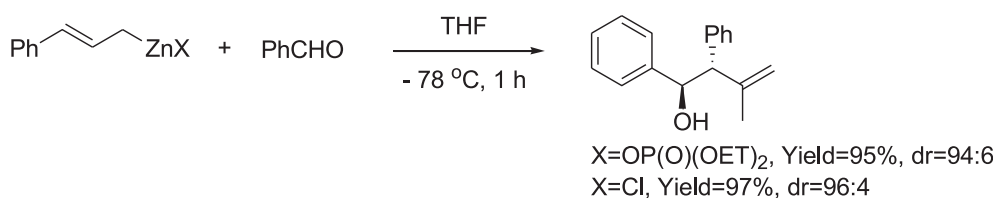
The allylzinc reagent was found to react with imines as well. Accordingly, the reaction of cinnamylzinc reagent with *N*-benzylidene-4-methyl-benzenesulfonamide in THF at -78°C afforded the corresponding *syn*-amine within 1 h in moderate yield with good diastereoselectivity. (Scheme 37) [102].



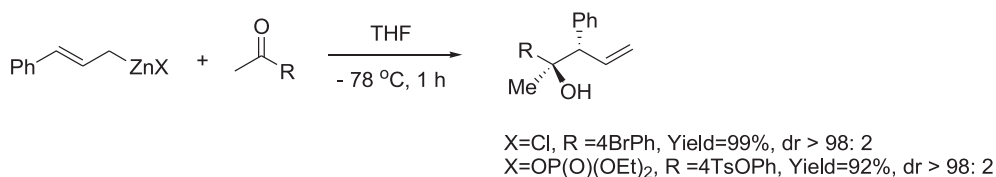
Scheme 29. Preparation and use of substituted allylzinc chlorides



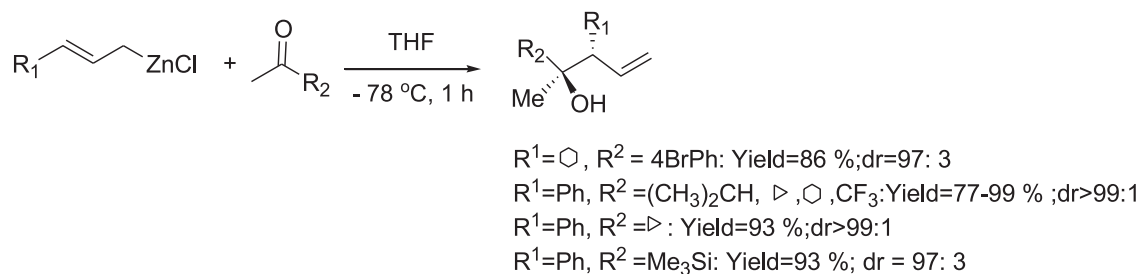
Scheme 30. Preparation of substituted allylzinc phosphates



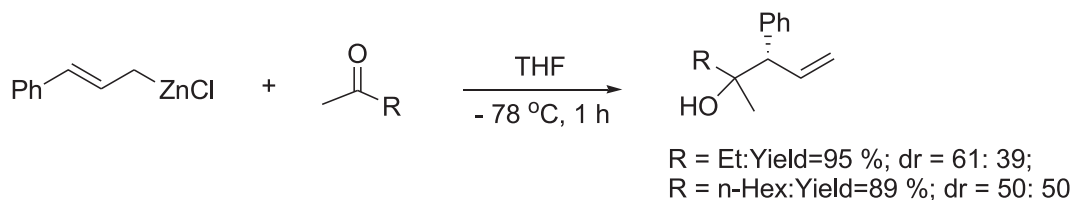
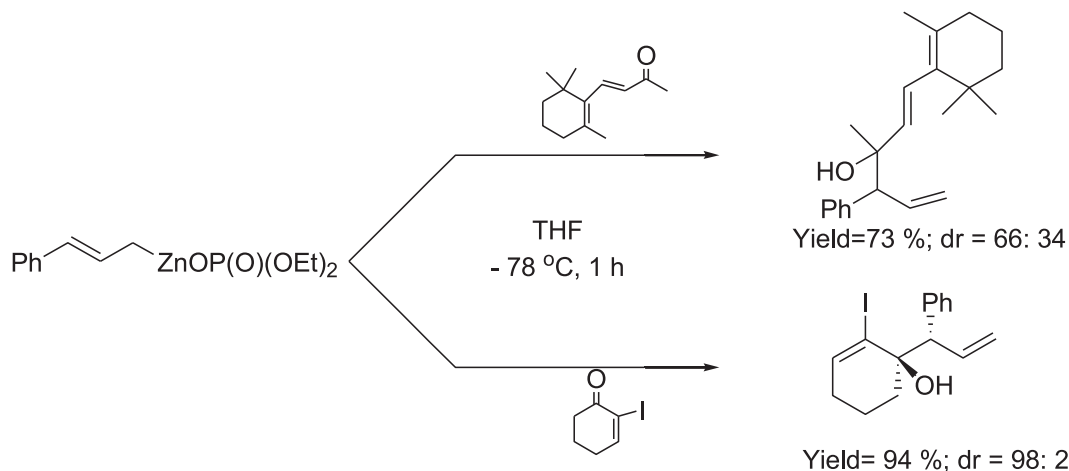
Scheme 31. Reactions of allylzinc chlorides or phosphates with benzaldehyde



Scheme 32. Reactions of allylzinc reagents with substituted acetophenones



Scheme 33. Synthesis of quaternary alcohols

**Scheme 34.** Reactions of cinnamylzinc reagent with linear methyl alkyl ketones**Scheme 35.** Reactions of alkyl vinyl ketones with the cinnamylzinc reagents

An efficient Zn-mediated diastereoselective allylation of chiral *N*-*tert*-butanesulfinyl imines at room temperature has been achieved for the synthesis of chiral homoallyl amines (Scheme 38). Most remarkably, the method allows the achievement of an opposite stereocontrol by simple tuning of the reaction conditions. In either case, the desired diastereomers were isolated with excellent diastereoselectivity (up to 98% de). With *N*-sulfinyl ketimines, the corresponding chiral homoallyl amines could also be obtained [103].

Allylzinc reagents, prepared by the LiCl-mediated direct insertion of zinc to allyl chlorides or phosphates, diastereoselectively react with carbonyl derivatives. To examine the possibility of controlling three adjacent stereocentres via this allylmatal addition, it was thought about to study the condensation of allylzinc reagents with a variety of α -substituted ketones. As cyclic ketones constitute more rigid systems, it was decided to study with α -substituted cyclohexanones. Thus, cinnamylzinc phosphate reacted with 2-methyl-cyclohexanone to afford the homoallyl alcohol in good yield as an only diastereomer (Scheme 39) [98,104].

Then this reaction was extended to several α -substituted cyclohexanones. Consequently, α -methoxy cyclohexanone reacted smoothly giving to the desired homoallyl alcohol in good yield and excellent diastereoselectivities within 1 h at -78°C . Similarly, larger substituents like α -acetoxy or α -benzyloxy group led to the

corresponding homoallyl alcohols in good yields and diastereoselectivities. Under the same conditions, 3-(2-oxo-cyclohexyl)-propionitrile reacted with the cinnamylzinc reagent to obtained the corresponding homoallyl alcohol in good yield and excellent diastereoselectivities within 1 h at -78°C (Scheme 40) [98,102].

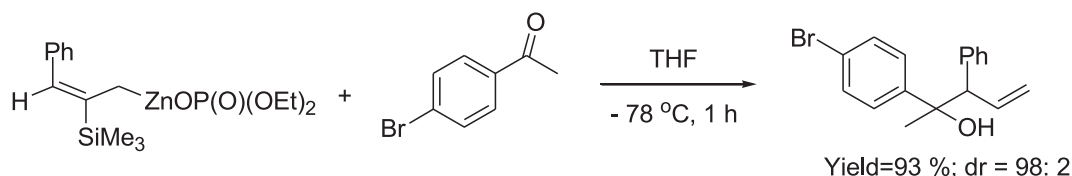
In the same way, other heteroatom substituted cyclohexanones was found to easily react with allylzinc reagents. Thus, both the chloro- and thiophenyl substituted homoallyl alcohols were obtained in good yield and diastereoselectivities (Scheme 41) [98,102].

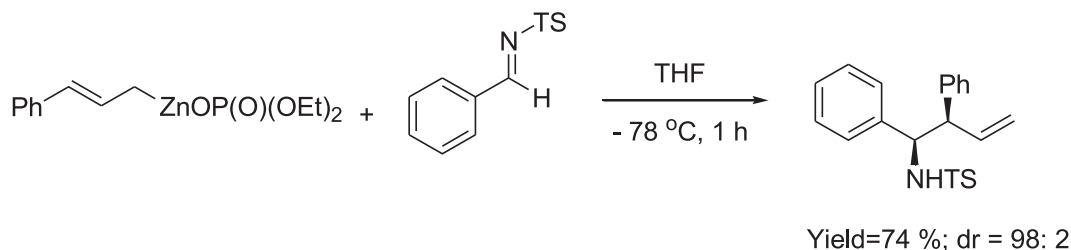
Interestingly, when 2-phenyl-cyclohexanone treated with cinnamylzinc phosphate under the same conditions, the corresponding homoallyl alcohol could only be obtained in 26% yield with good diastereoselectivity. This may be due to the competitive deprotonation of the benzylic proton α to the keto function (Scheme 42) [98].

Noteworthy, 2-methyl-cyclopentanone leads to the homoallyl alcohol in good yield as a single diastereoisomer when treated with cinnamylzinc phosphate. (Scheme 43) [98].

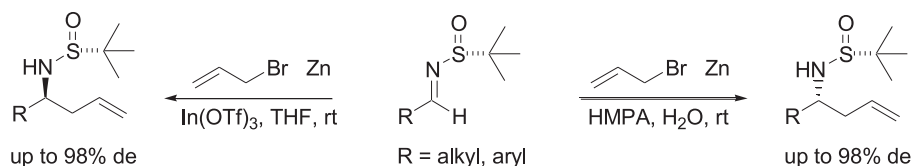
The diastereoselectivity was found to decrease when the substituted allylzinc reagent was treated with 2-methoxy-cyclohexanone, to afford the corresponding homoallyl alcohol in good yield with the mixture of two diastereomers (Scheme 44) [98].

After all, it is attractive to observe that this addition is only

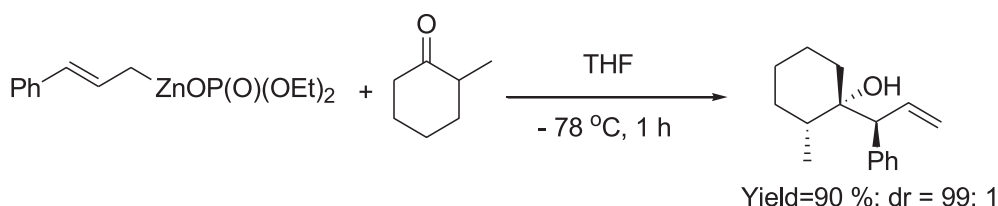
**Scheme 36.** Reaction of bulky trimethylsilyl-substituted allylic zinc reagent



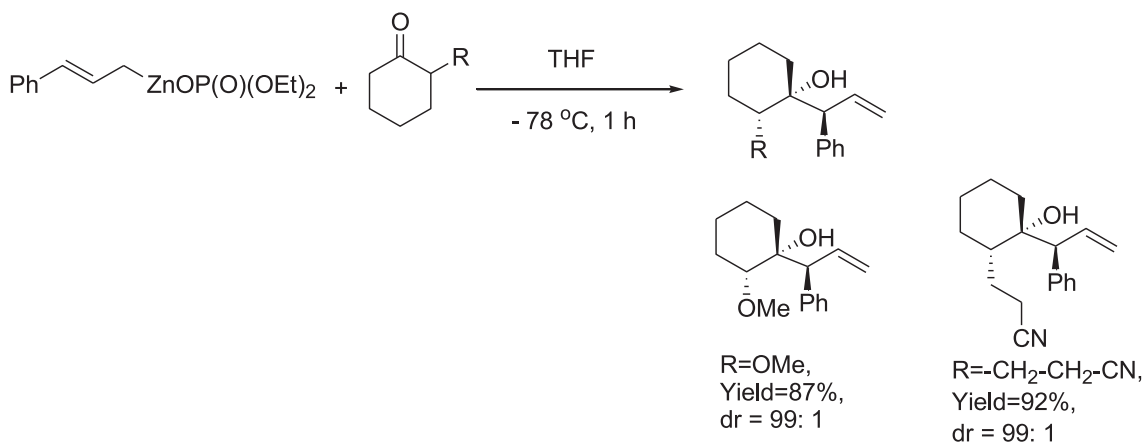
Scheme 37. Reaction of cinnamylzinc phosphate reagent with imines



Scheme 38. Zn-mediated diastereoselective allylation of chiral N-tert-banesulfinyl imines



Scheme 39. Reaction of cinnamylzinc phosphate with 2-methyl-cyclohexanone



Scheme 40. Reactions of cinnamylzinc phosphate with α -substituted cyclohexanones

selective with α -substituted ketones; definitely, when cinnamylzinc chloride was reacted with 3-methyl cyclohexanone, only a 50:50 mixture of diastereomers could be obtained. The selectivity observed in this reaction can be rationalised by considering a cyclic chair-like transition state, where the allylzinc reagent approaches from the sterically less hindered side [96,98] (Fig. 1).

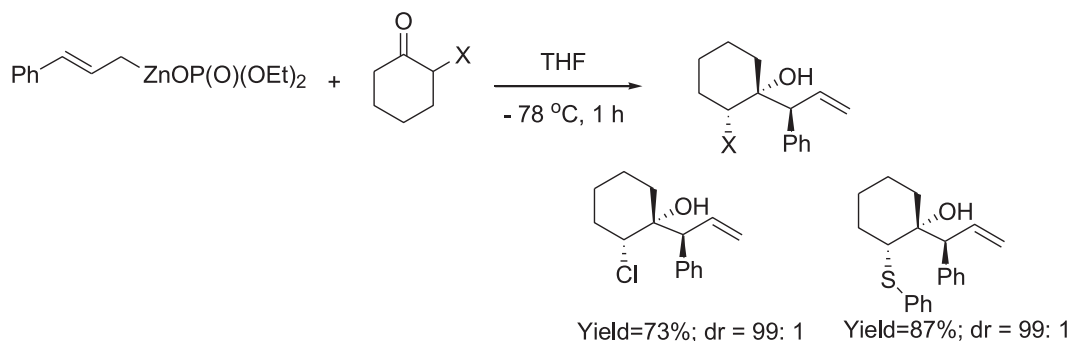
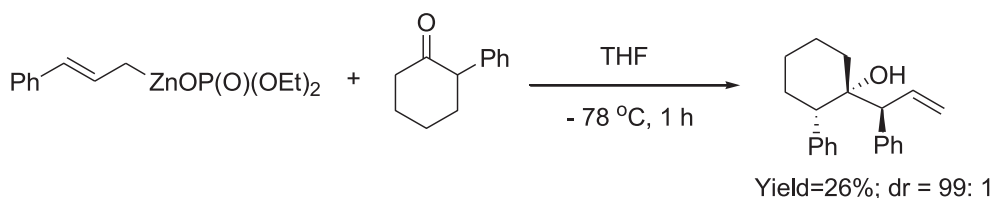
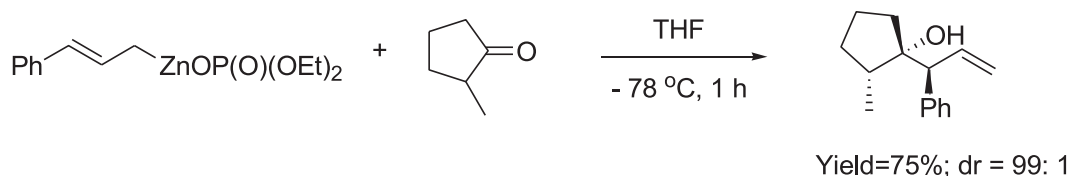
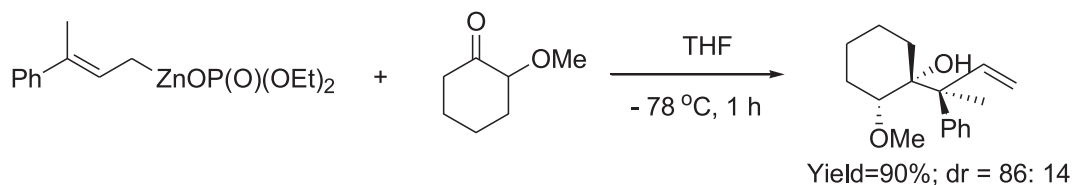
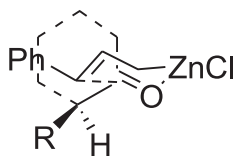
When 2-allyl-cyclohexanone was treated with cinnamylzinc phosphate, the corresponding homoallyl alcohol was isolated in good yield, as a single diastereoisomer (Scheme 45) [98].

The reaction of allylzinc reagents with acyclic α -substituted ketones was then studied. Thus, cinnamylzinc chloride was reacted with 3-chloro-butan-2-one in THF at -78°C . After 1 h, the

homoallyl alcohol was obtained in good yield as a single diastereoisomer (Scheme 46) [98].

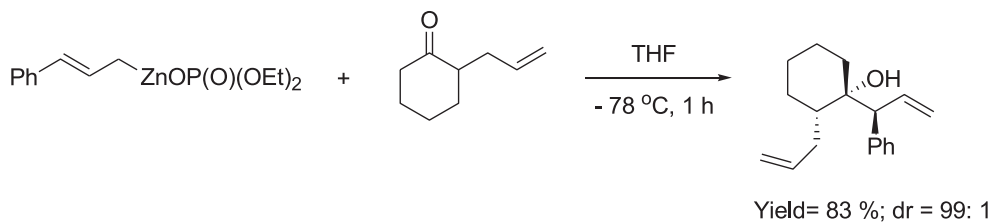
Other 3-halo-butan-2-ones treated with cinnamylzinc chloride under the same reaction conditions. Remarkably, instead of the corresponding homoallyl alcohols, epoxide was isolated in good yield (Scheme 47). Similarly, 3-tosyloxy-butan-2-one led to epoxide, when treated with cinnamylzinc chloride. In this case, the observed selectivity can be explained considering the Cornforth model [98,104].

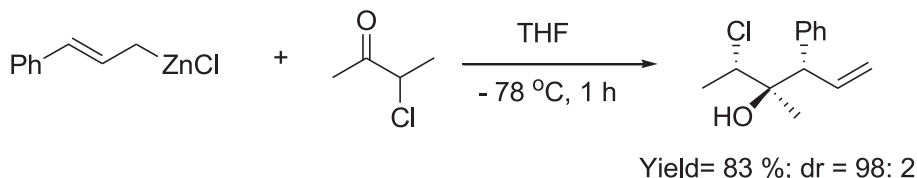
When non-halide groups like -Ph or -SPh were used as substituents, the reaction produced the corresponding homoallyl alcohols in excellent yields, but with lower selectivity (Scheme 48)

**Scheme 41.** Reactions of cinnamylzinc phosphate with heteroatom substituted cyclohexanones**Scheme 42.** Reaction of cinnamylzinc phosphate with 2-phenyl-cyclohexanone**Scheme 43.** Reaction of cinnamylzinc phosphate with 2-methyl-cyclopentanone**Scheme 44.** Reaction of substituted allylzinc reagent with 2-methoxy-cyclohexanone**Fig. 1.** Chair-like transition state.

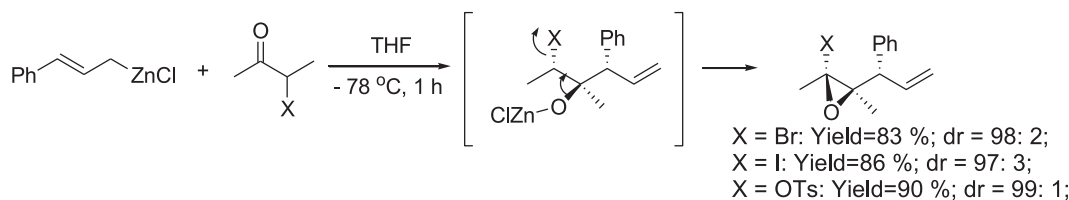
[98]. Thus, the phenyl substituted homoallyl alcohol product was obtained in good yield with a moderate diastereoselectivity. Under the same conditions, the thiophenyl-alcohol was isolated in almost quantitative yield, as a mixture of two diastereomers.

A variety of polysubstituted δ -lactones containing three or four stereocenters were synthesized from a variety of dialkyl 2-(3-oxo-1,3-diarylpropyl)malonates by a Barbier-type zinc-mediated

**Scheme 45.** Reaction of cinnamylzinc phosphate with allyl substituted cyclohexanone



Scheme 46. Reaction of cinnamylzinc reagents with acyclic α -halo ketone



Scheme 47. Reactions of cinnamylzinc reagents with acyclic α -substituted ketones

allylation or cyclohexenylation of the keto group, followed by intramolecular lactonization/transesterification (Scheme 49) [105].

2.4. Green allylation by allylzinc species

Allylation of carbonyl compound with allylic organometallic reagents are a well known process with a broad spectrum of synthetic applications. Electrochemical methods are an excellent approach to synthesis because they can eliminate (or dispose of) waste materials and recycle metal reagents. Again utilization of aqueous reaction also will be an excellent green approach. As a part of the rapidly emerging field of green chemistry, developing more benign synthetic procedures in organic synthesis is attractive in going towards sustainable approaches [17]. Various approaches are accessible including avoid of the utilization of organic solvents as the reaction media to reduce the amount of waste, energy usage, and to avoid volatile, toxic and flammable solvents.

In 1993 Perichon and co workers. reported a simple and mild method for zinc activation using an electrochemical process, based on the cathodic reduction of a catalytic amount of zinc bromide in the presence of a zinc anode. This process has been applied to the coupling of carbonyls with allylic bromides to afford the corresponding homoallyl alcohols (Scheme 50) [106].

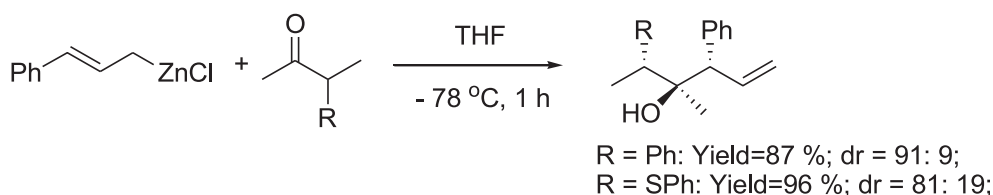
Electroallylation reaction is achieved in aqueous ammonia in an undivided cell fitted with a pair of Zn electrodes. In this novel procedure the reaction proceeds in an aqueous alkaline medium and a series of functional groups was found to tolerate in such mild conditions. The reaction conditions involve the use of 0.5 mmol aldehyde, 1 mmol allyl bromide, 5 mL 4.5 M aqueous ammonia, $I = 15$ mA, 1.5 cm² zinc electrodes, H₂O:THF (4 : 1) as solvent at room temperature (Fig. 2, Scheme 51). When allyl bromide was replaced with crotyl bromide the regioselectivity of anti:syn = 41 : 59 was obtained. Zn foils were chosen as both anode and cathode in this one-pot electrochemical process [107].

Huang et al. in 2010 reported another facile electrochemical allylation reaction of carbonyls with a consumption of catalytic amount of metallic zinc in aqueous electrolyte in a divided cell. Here Nafions-551 a cation-exchange membrane separates cathode and anode chambers and the carbonyl and allyl bromide were added into the cathode chamber. When H₂SO₄ (0.1 N) was used as an electrolyte, the yield of the reaction was 83% with a catalytic consumption of zinc. The consumption of zinc was reduced in alkaline solutions (4.5 M aq. NH₃) and neutral (0.1 M LiClO₄) solutions however the yield was very lower (Scheme 52) [108].

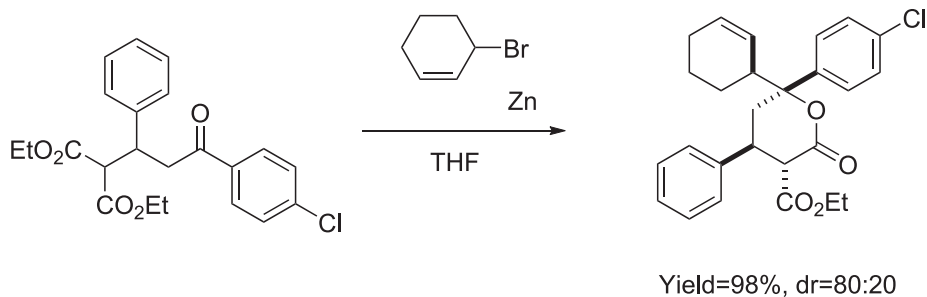
Aromatic aldoxime esters were found to give facile Zn-mediated Barbier-type allylation reaction affording the corresponding protected homoallylic hydroxylamines, which were readily transformed into homoallylic hydroxamic acids. The advantage of this reaction is that it is effective with non-activated aromatic aldoximes. The products are readily converted into homoallylic hydroxamic acids, which are valuable compounds both in organic synthesis and medicinal chemistry [109]. This tandem reaction provides a convenient procedure to afford hydroxamic acids from easily available aromatic aldoxime esters (Scheme 53).

A very fast and efficient zinc-mediated Barbier-type reaction that allows the synthesis of homoallyl alcohols at room temperature from carbonyls with allyl halides under solvent-free conditions has been achieved. The scope of this reaction was further extended to a variety of ketones with allyl bromide mediated by zinc powder under solvent-free conditions. The synthetic applications of organozinc reagents on developing rapid and novel green chemistry reactions have been demonstrated (Scheme 54) [110].

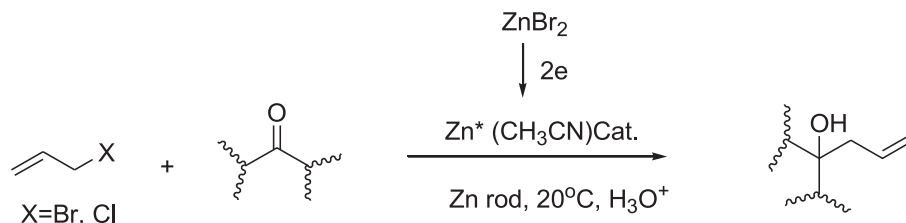
Tagliavini and co-workers reported a Zn powder mediated very versatile procedure for the synthesis of homoallyl alcohols. The stereochemical course of the allylation of several aldehydes with crotyl halides using Zn powder as the condensing agent in water/cosolvent (salt) media has been widely studied. Radical ions of well-defined geometry such as $[\text{CH}_2=\text{CHCH}(\text{CH}_3)\text{X}]^-$ (X = Cl or Br)



Scheme 48. Reactions of cinnamylzinc reagents with acyclic α -substituted ketones



Scheme 49. Allylation or cyclohexenylation of the keto group, followed by intramolecular lactonization/transesterification



Scheme 50. Coupling of carbonyls with allylic bromides by electrochemical zinc activation

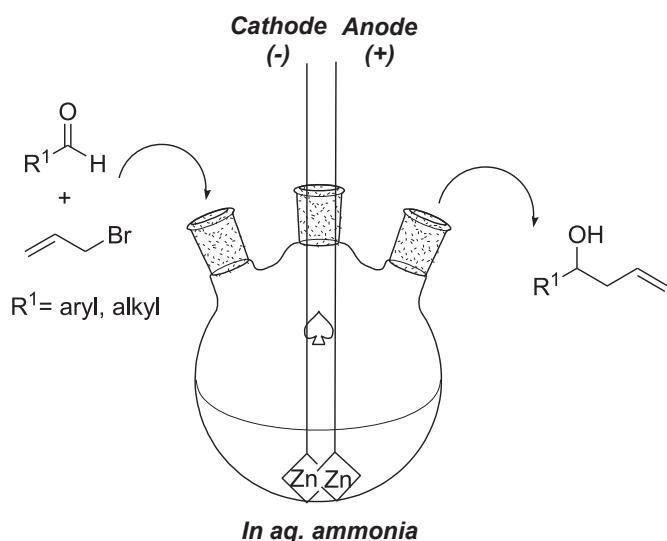
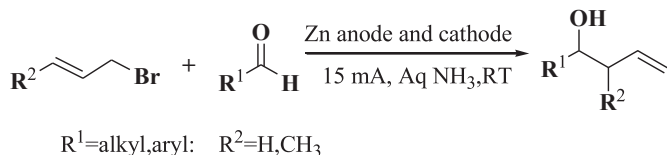
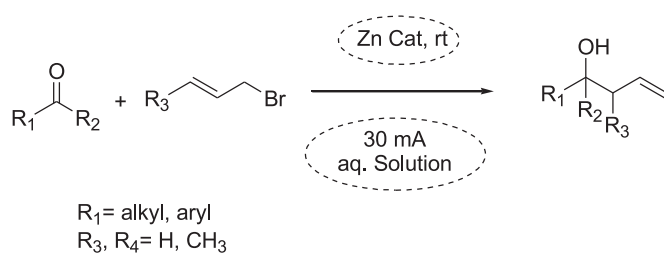


Fig. 2. Electrochemical allylation in aqueous ammonia.

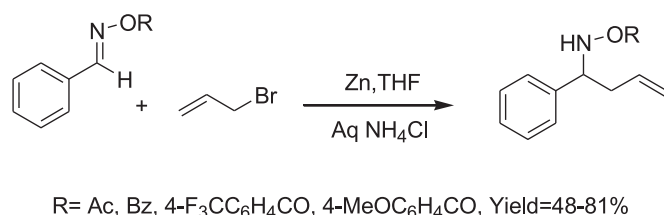


Scheme 51. Zn-mediated electrochemical allylation of aldehydes in an aqueous ammonia solution

are thought to be produced *via* an electron-transfer process, which seems to be accelerated by the high interfacial energy developed by the water. These radicals were trapped by the aldehyde molecules to afford isomeric mixtures of *threo-erythro* α -methylhomoallyl alcohols. The stereochemistry seems to be determined by the structure of the radical anions adsorbed on the Zn particles and is



Scheme 52. Electrochemical allylation reaction of carbonyls with catalytic zinc

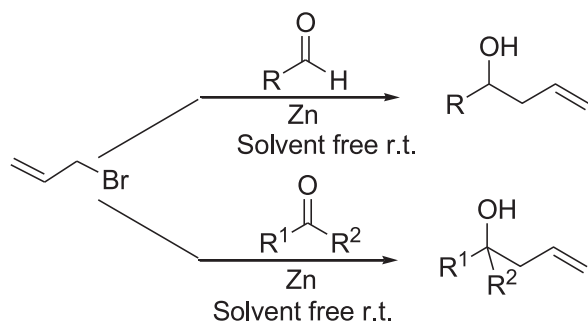


Scheme 53. Zn-mediated Barbier-type allylation reaction of aromatic aldoxime esters

not dependent on the geometry of the organic halides, the co-solvents, or the salts. Reactions performed in the presence of Bu_3SnCl show no discrimination between the coupling reactions of the allyl bromide with aldehydes *i.e.* in presence and absence of Bu_3SnCl (Scheme 55) [111].

Jing et al. reported another Zn-mediated electrochemical allylation reaction of simple imines in a mixture of saturated aqueous NH_4Br and THF (9:1) at room temperature. The reaction conditions involved 0.5 mmol imine, 1 mmol allyl bromide, saturated aqueous NH_4Br with THF (9:1) as electrolyte (6 ml), constant current of 30 mA and 1.5 cm^2 Zn electrodes (Scheme 56) [112].

By using an efficient, sustainable and green approach, the waste Zn material in Zn(0) promoted Grignard–Barbier type allylation of aldehydes has been successfully applied by prathihar et al. as a reusable material for the adsorption of a variety of dyes and also converted it into the corresponding hexagonal wurtzite phase of



Scheme 54. Zinc-mediated Barbier-type allylation reaction under solvent-free conditions

ZnO. The conversion of the corresponding Zn(0) to its hydroxide or ZnO was investigated by XRD analysis, which confirmed the facile conversion of Zn(OH)₂ to its corresponding ZnO after calcination at 300 °C. The reported procedure is a cost-effective chemical path for producing the ZnO nano material at the gram level. The Zn waste obtained from allylation reaction shows an excellent ability as a reusable material for the adsorption of Congo red, methyl red and methylene blue, and is expected to be useful in many other applications. The easy synthesis, effortless separation, efficient recycling and good adsorption capability of the waste Zn nano material toward several dye molecules make it a competent applicant for waste water treatment (Fig. 3) [113].

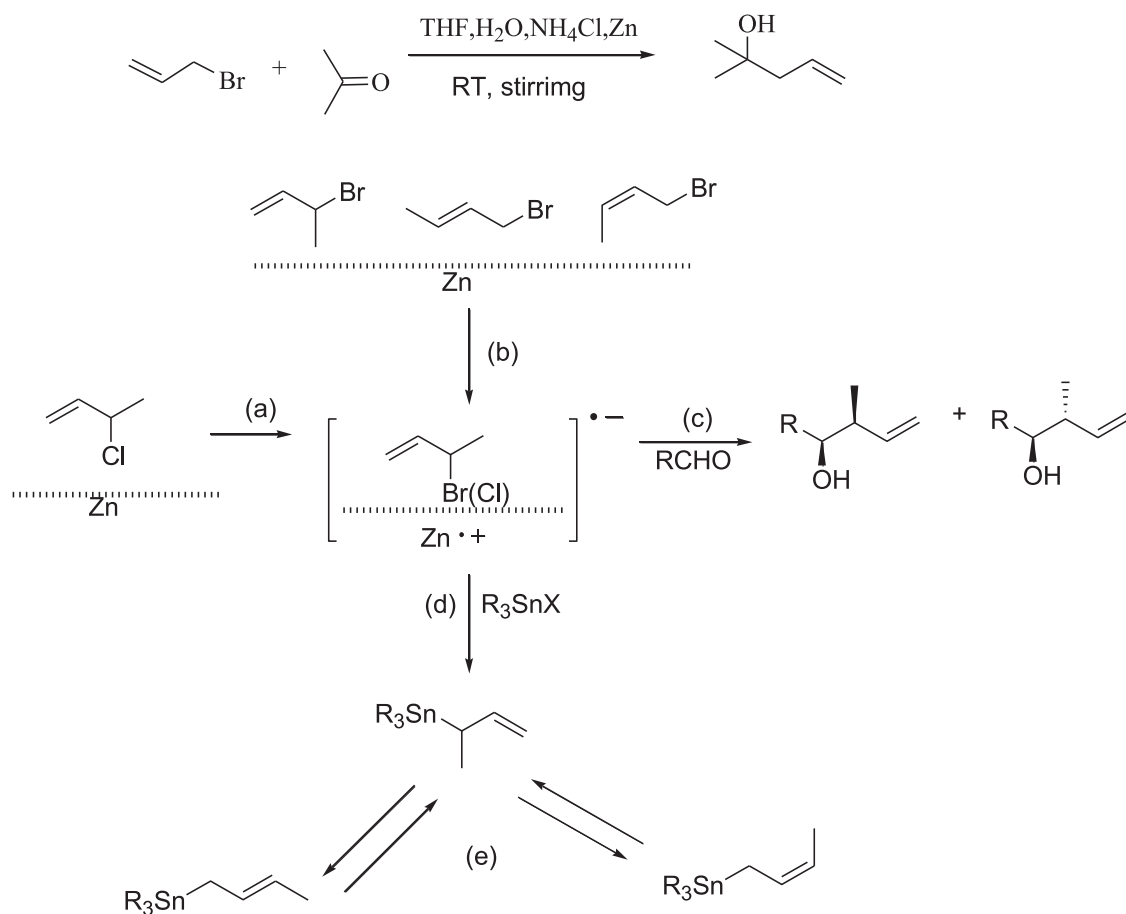
Roy and co-workers reported a new aqueous electrochemical

process for allylation reactions using zinc chloride as the reagent. A unit galvanic cell using a beaker containing an aqueous zinc chloride solution, zinc rod, TCO glass plate and connecting copper wire has been designed. Instead of using a connecting copper wire, an external bias can be used for applying voltage from the outside. The zinc rod served as a self-decaying anode and the TCO glass substrate as the cathode. During the process, wire-shaped nano zinc architecture was formed from zinc chloride on the cathode. Wire-shaped nano zinc is the active reagent for the allylation of aldehydes or imines. The corresponding homoallylic alcohols or amines are prepared in good yields. Both the zinc salts and water can be recycled (Fig. 4, Scheme 57) [114].

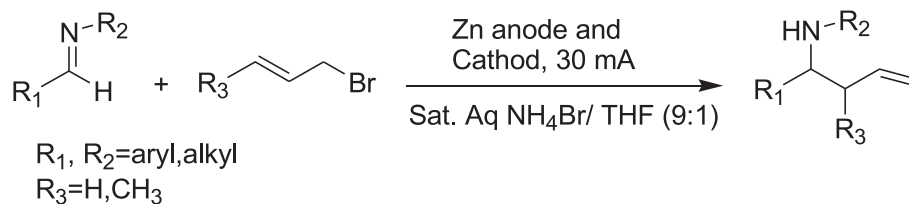
3. Propargylic zinc reagents

3.1. Introduction

Homopropargyl and α -allenyl alcohols and amines are very important precursors for the synthesis of bio-active molecules. In comparison to a number of studies on the reactivity of allylmetal reagents towards aldehydes, ketones, epoxides or imines, fewer studies are known for the addition of propargyl or allenyl metallic reagents to various aldehydes, ketones, epoxides or imines. One may note that the reaction offers exciting challenges since various selectivity and reactivity issues are involved which include metal-lotropic rearrangement of the initially produced propargyl organometallics to the corresponding allenyl equivalent and subsequent reactions leading to the corresponding alcohols for which the product distribution is determined by the position of the



Scheme 55. Stereochemical allylation and crotylation of aldehydes using Zn powder



Scheme 56. Zn-mediated electrochemical allylation reaction of simple imines

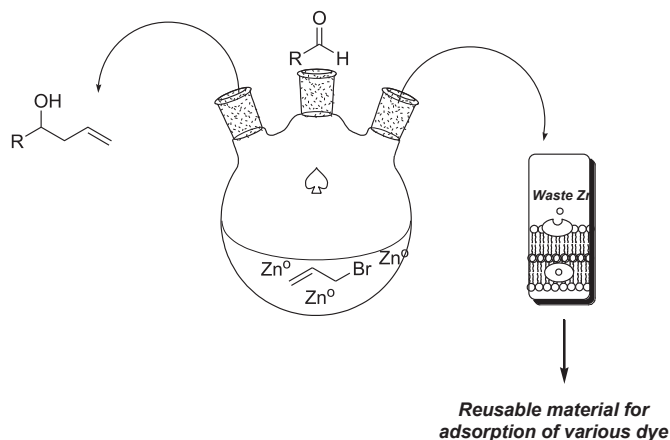


Fig. 3. Reusable material for various dye.

equilibrium between the two organometallic intermediates [115,116].

3.2. Generation and reactivity of propargylzinc

In 2005 Jin-Xian Wang et al. reported a rapid convenient and efficient procedure for the Zn-mediated Barbier-type propargylation reaction of carbonyls. They achieved a solvent-free synthesis of homopropargyl alcohols at room temperature with no side reactions for example reduction or self-coupling. Focusing their attempts on developing rapid and novel green chemistry reactions they explored the synthetic applications of propargylic organozinc reagents. Such solvent-free synthesis provided a rapid, convenient and efficient method for the synthesis of homopropargylic alcohols by coupling aldehydes or ketones with propargyl bromides mediated by Zn powder. Most significantly, the

reported reaction conditions achieved the advantages of excellent yields (78–87%), shorter reaction time (10–30 min), environmentally benign reaction conditions and its versatile applicability to both aldehydes and ketones (Scheme 58) [110].

In 2009 Xiaofang Maa et al. with the huge interest and demands of sustainable green chemistry reported another Barbier-type propargylation reaction of carbonyl compounds with propargyl bromides by reactive Zn–Cu couple mediator under solvent-free reaction conditions to produce the homopropargyl alcohols as the only product in excellent yields at room temperature without the formation of allenyl alcohols (Scheme 59). The ketones also found to react with propargyl bromides to produce the corresponding homopropargylic alcohols in good to excellent yields at –14 to –16 °C. This method has several advantages of excellent yields, shorter reaction time, high regioselectivity, and exclusion of the use of organic solvents (Tables 1 and 2). The Zn–Cu couple has been considered as a moderately less-toxic, very cheap metal mediator in organic reactions, and has various advantages such as simple post-processing or work up, complete recovery of Cu, and environment friendly process [117].

Bieber et al. examined the more reactivity aspects of Zn metal. Zinc is traditionally the earliest metal used for the preparation of isolated, stable organometallic compounds and for synthetically valuable carbon–carbon bond formation reactions, for its impressive outcomes similar to aqueous Barbier-type processes with allyl halides and bromo esters. *In situ* propargylzinc halide or alkyl substituted propargylzinc halide (Cl or Br) reacted with aromatic, hetero aromatic, unsaturated and aliphatic aldehydes, ketones in different aqueous concentrated salt solution to afford highly selective homopropargyl alcohols similar to the addition of magnesium, and aluminium reagents under anhydrous conditions (Scheme 60) [118].

In 2010 Daniel R. Fandrick et al. reported the Zn catalyzed and mediated asymmetric propargylation of carbonyl compounds with propargyl borolanes which involves the reaction of catalytic

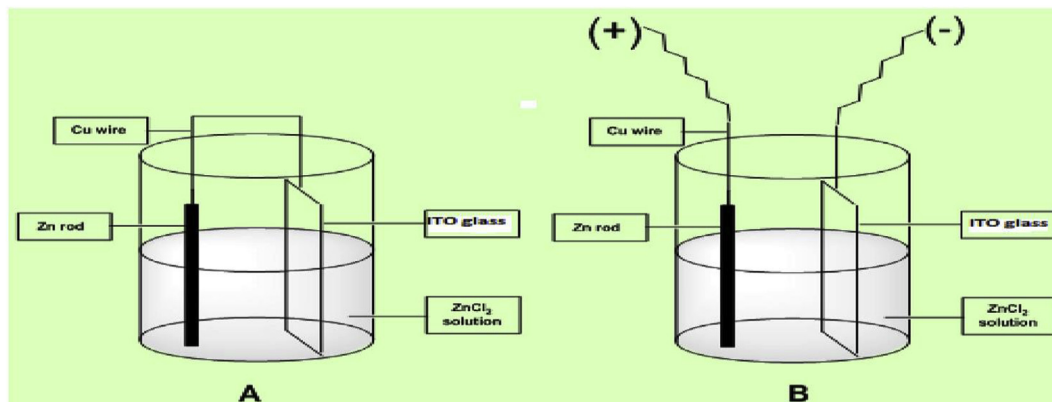
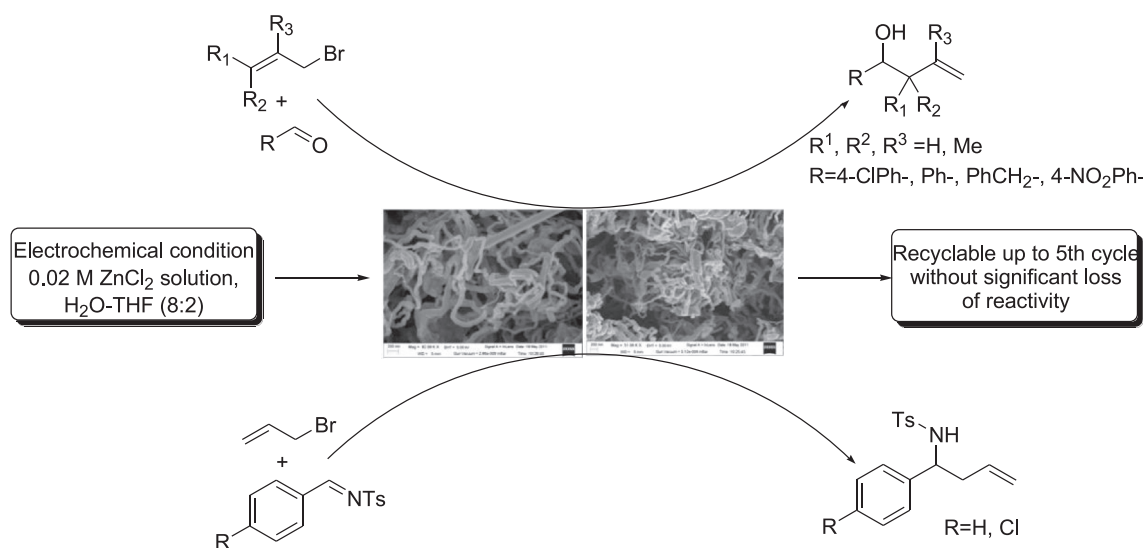
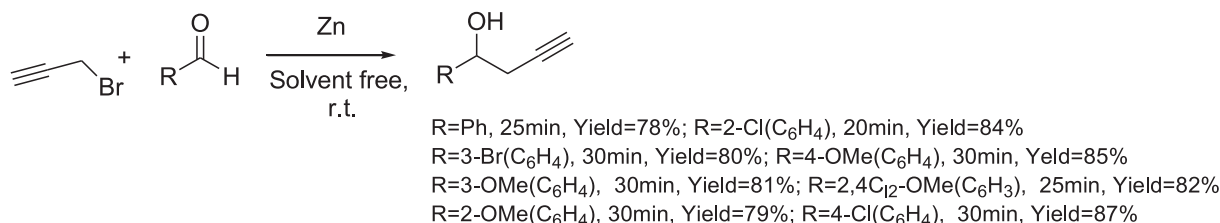


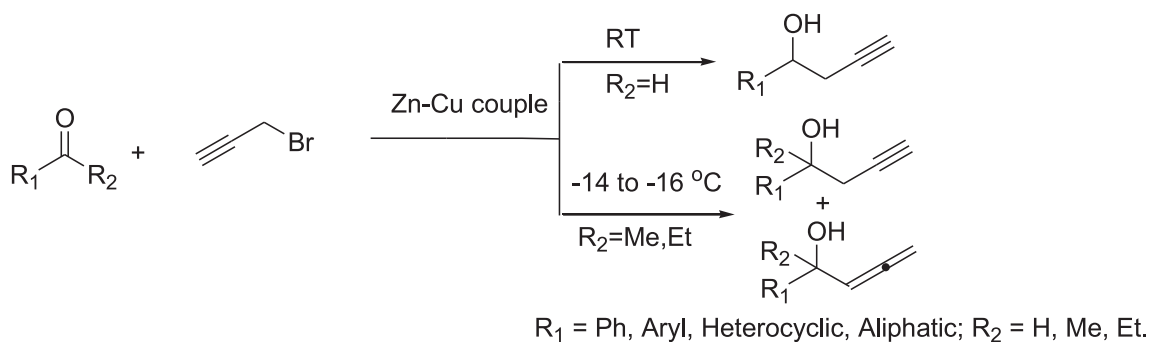
Fig. 4. Electrochemical set up for allylation and propargylation



Scheme 57. Electrochemical synthesis of nano-zinc wire and its Reactivity in allylation of aldehydes



Scheme 58. Zn-mediated solvent-free Barbier-type propargylation reactions



Scheme 59. Zn–Cu couple mediated solvent-free Barbier-type propargylation reactions

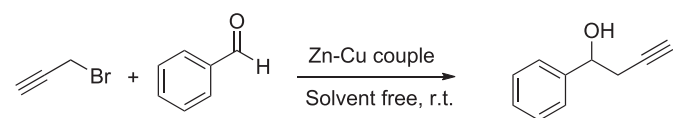
amount of Zn with propargyl or allenyl borolanes. propargyl diethanolamine borolanes. They proposed a catalytic cycle for the overall process. This Zn mediated propargylation reaction showed wide substrate scope, and both the propargyl and allenyl borolanes can be utilized (Scheme 61). The propargylation was also achieved with air-stable and crystalline air-stable and crystalline diethanolamine propargyl boronates [46].

Daniel R. Fandrick et al. presented zinc-mediated and catalyzed moderate to high stereoselective asymmetric propargylations of both aliphatic and aromatic trifluoromethyl ketones with a propargyl borolane and the chiral *N*-isopropyl-*L*-proline ligand. The provided methodology was successfully applied for the synthesis of the Glucocorticoid agonist BI 653048 on a multi-kilogram scale. Catalytic amount of water was found to accelerate the Zn-catalyzed

propargylation reaction with a prominent effect on the stereo-selective reaction with trifluoromethyl ketones (Scheme 62) [119].

Tagliavini and co-workers investigated and widely studied the stereochemical track of the propargylation reaction of various aldehydes with propargyl bromides using Zn powder as the condensing mediator in various solvents. Radical ions of well-defined geometry are thought to be produced via an electron-transfer process, which seems to be accelerated by the high interfacial energy developed by the water. They adopted a Zn powder mediated very versatile procedure for the synthesis of homo-propargyl alcohols. The stereochemistry was determined by the structure of the radical anions adsorbed on the Zn particles and was independent of the geometry of the organic halides, the co-solvents, or the salts (Scheme 63) [111].

Table 1
Effects of the mediator and solvents.



Entry	Mediator(s)	Solvent	Time	Yield(%)
1.	Zn–Cu couple.	Water	8 h	Trace
2.	Zn–Cu couple.	Water	16 h	20
3.	Zn–Cu couple.	Diethylether	16 h	25
4.	Zn–Cu couple.	Dichloromethane	16 h	40
5.	Zn–Cu couple.	Acetonitrile	3 h	42
6.	Zn–Cu couple.	Ethanol	6 h	54
7.	Zn–Cu couple.	Dimethylsulfoxide	1.2 h	60
8.	Zn–Cu couple.	Tetrahydrofuran	3 h	75
9.	Zn–Cu couple.	Dimethylformamide	1.2 h	70
10.	Zn–Cu couple.	Solvent free	0.5 h	85
11.	M ¹ –Cu.	Solvent free	16 h	No result
12.	Sn–Cu.	Solvent free	1.2 h	56
13.	Mg–Cu.	Solvent free	1.5 h	12
14.	M ² .	Solvent free	5–17 h	No result
15.	Sn.	Solvent free	1 h	60
16.	Zn.	Solvent free	0.42 h	78

Reaction conditions: Carbonyl (2.0 mmol), allyl bromide (3 mmol), Zn–Cu couple or metal (M) (4 mmol), solvent (2 ml), at room temperature, M¹ = Al, Fe, Mn, or Cd, M² = Cd, Ni, Mn, Co, Fe, Al, W, Pb or Ti.

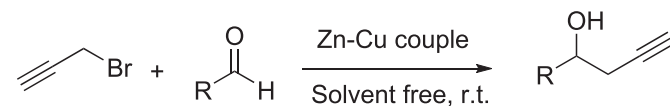
In 2001 Artur Jogi and Uno Maeorg represented a facile route for the Zn power mediated Barbier-type propargylation reaction of carbonyls with both substituted and unsubstituted propargyl bromides in a mixture of saturated aq. NH₄Cl-THF solution to achieve the selective synthesis of corresponding propargyl and allenic alcohols. This was the first successful synthesis of these propargylic and allenic alcohols with cheap Zn powder and the first reported synthesis of the allenic alcohol 3-hexyl-1,2- heptadien-4-ol. They also explained the regiospecific Barbier-type reaction of two representative model compounds producing the corresponding substituted propargyl or allenic alcohol as exclusive product. The coupling of substituted and unsubstituted propargyl bromide with butanal using Zn powder afforded the corresponding propargylic and allenic alcohols in modest yields with high selectivity (Scheme 64) [120].

In 2011 Zhang et al. developed a simple and novel method for the zinc-mediated propargylation reactions of various aldimines with propargyl bromide for the synthesis of the corresponding homopropargyl amines under solvent-free and catalyst-free conditions. This methodology for the activated zinc powder mediated propargylation of aryl aldimines involves a short reaction time (5 min) at room temperature. The reaction here has the following advantages: mild conditions, short reaction times, high yields and flexibility to a broad range of imines (Scheme 65) [121].

Yanagisawa reported that Grignard type propargylzinc reagent gave regioselective propargylation in tetrahydrofuran using acylsilane or aldehyde as electrophile. The novel method describes the synthesis of homopropargylic alcohols using acylsilanes as the electrophiles. The propargylzinc reagent, generated from 2-octynyl bromide and zinc dust in THF, treated with benzoyltrimethylsilane followed by desilylation achieved the corresponding homopropargyl alcohol as the only single product with no contamination of any regio-isomer (Scheme 66). In contrast with benzaldehyde, a 35:65 mixture of acetylenic and allenic alcohol was obtained under similar reaction conditions with the same zinc reagent (Table 3). This methodology was also applicable for synthesis of PGE₃ and F_{3α} ester [122].

In 2015 Lasch et al. reported the propargylation of *tert*-butyl 2-

Table 2
Zn–Cu couple mediated regioselective propargylation of various aldehydes with propargyl bromide.

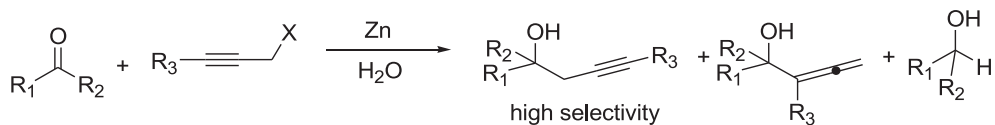
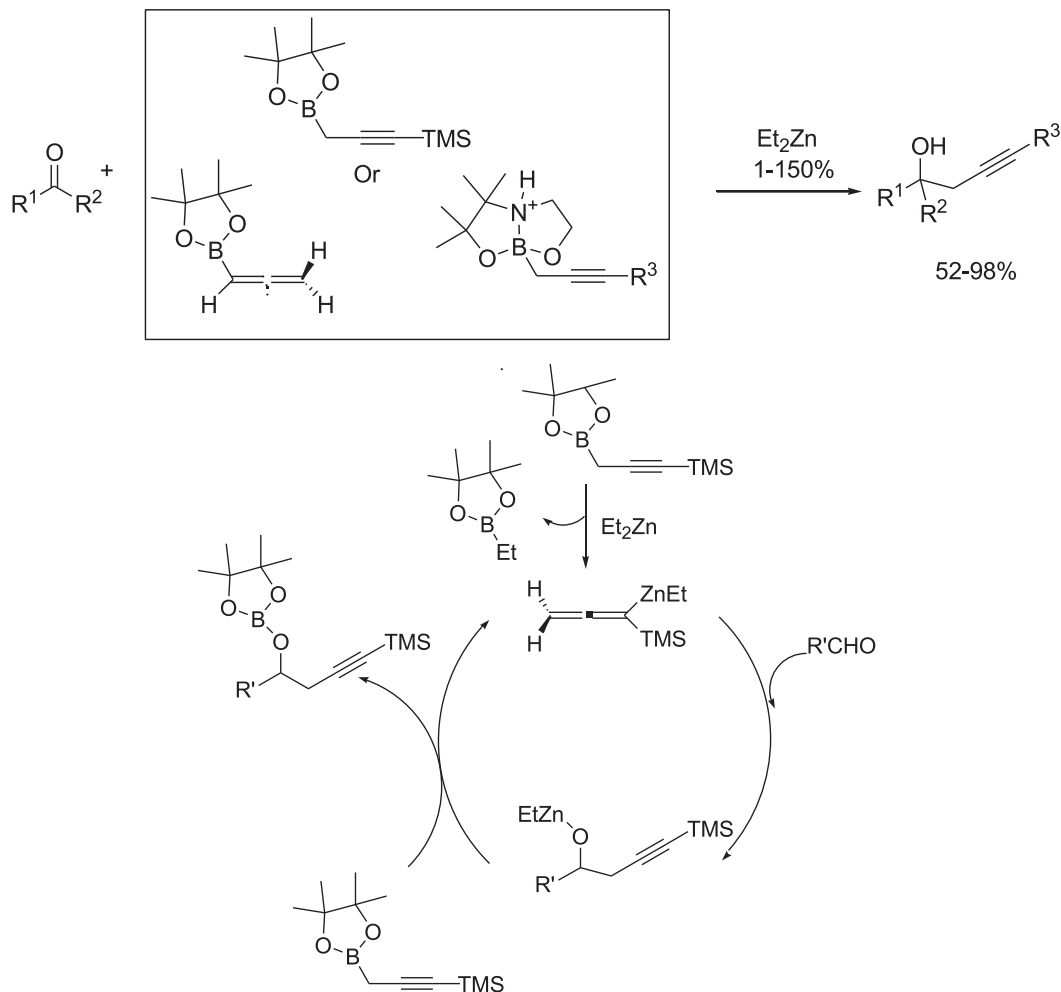
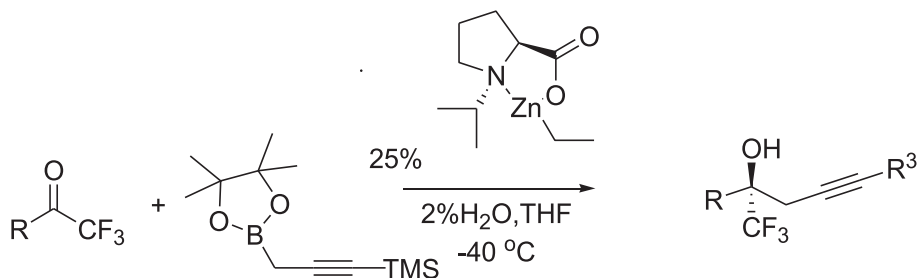


Sl. No.	R or aldehyde	Time(min)	Yield(%)
1	Ph	30	85
2	4-OMe(C ₆ H ₄)	30	88
3	2-OMe(C ₆ H ₄)	30	86
4	3-OMe(C ₆ H ₄)	30	84
5	2,4Cl ₂ -OMe(C ₆ H ₃)	45	83
6	2,6Cl ₂ -OMe(C ₆ H ₃)	45	83
7	2-Cl(C ₆ H ₄)	45	90
8	3-Cl(C ₆ H ₄)	45	89
9	4-Cl(C ₆ H ₄)	45	89
10	3-Br(C ₆ H ₄)	25	88
11	4-Br(C ₆ H ₄)	25	84
12	2-Br(C ₆ H ₄)	25	83
13		30	80
14		30	90
15		30	90
16		55	85
17		30	84
18	(2-Bromo-vinyl)-benzene	30	88
19	3-(4-Methoxy-phenyl)-propenal(aldehyde)	30	80
20	Hexane	55	60

(4-bromophenyl)azocarboxylate ester with propargyl bromide, (80% in toluene) in tetrahydrofuran that is saturated with aq. NH₄Cl treated with Zn in 4 min. Two experiments with azocarboxylate and propargyl bromide established that the Barbier-type procedure, in which Zn powder was added to both, was obviously better for propargylation reactions, as the addition of pre-mixed Zn powder and propargylic bromide resulted desired hydrazine in only 28% yield compared to 61% for the Zn addition. In the propargylation, functionalization selectively appeared at the β-nitrogen atom of the azocarboxylic esters, which is adjacent to the aromatic core (Scheme 67) [123].

The Marshall type propargylation showed that the aldehydes were reductively coupled with propargyl mesylates to construct a new C–C bond. The transfer of chirality from the mesylates allows for reliable construction of adjacent stereogenic centers. Reagents such as diethylzinc and indium(I) iodide were used as reducing agents. This special type reaction has been established to be valuable in large scale synthesis. The overall mechanism involves generation of the chiral allenylzinc reagent involving redox cycle of the Pd catalyst. This allenylzinc reacts with the aldehyde via a cyclic TS with the transfer of chirality (Scheme 68) [124].

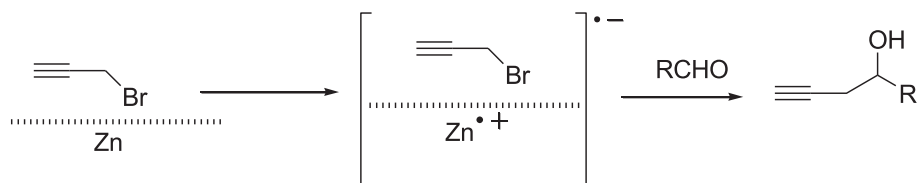
A Zn-catalyzed diastereoselective propargylation reaction of *t*-

**Scheme 60.** Selective propargylation reactions**Scheme 61.** Zn catalyzed and mediated asymmetric propargylation of carbonyl compounds with propargyl borolanes**Scheme 62.** Stereoselective, asymmetric propargylations of trifluoromethyl ketones with a propargyl borolane

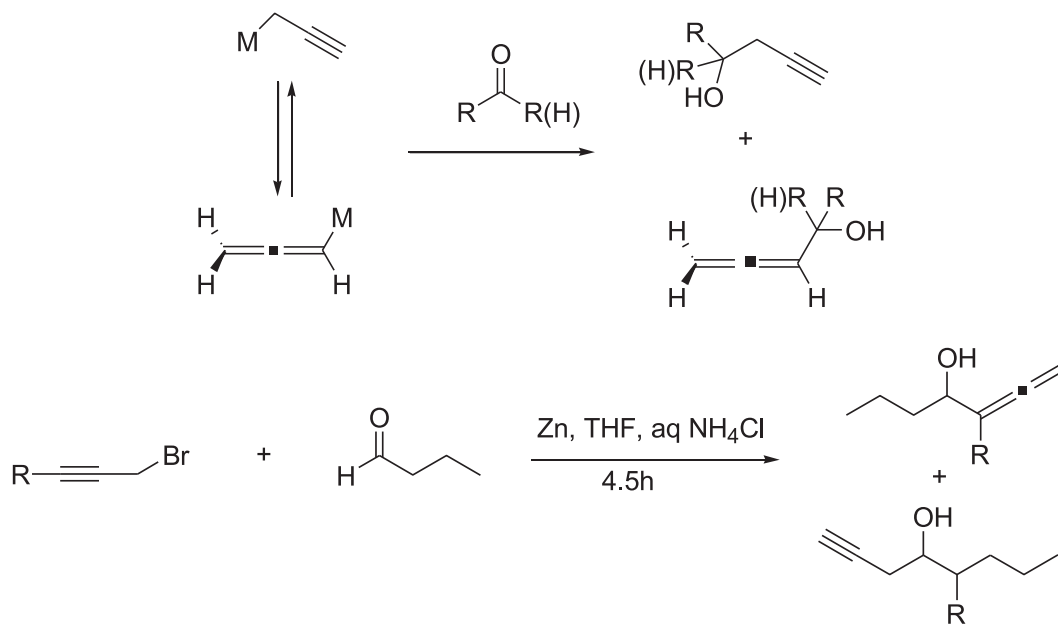
butanesulfinyl imines was reported. This method was used for the synthesis of both aromatic and aliphatic homopropargyl amines in up to 98:2 and 99.8:0.2 dr, respectively (Scheme 69). The utility of the homopropargyl amines was established by the application to

the synthesis of a *cis*-substituted pyrido-indole through a diastereoselective Pictet-Spengler cyclization [125].

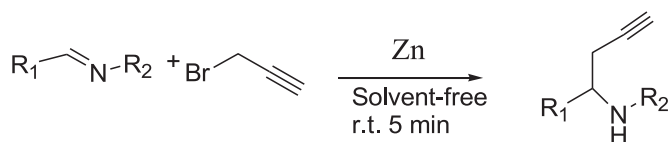
Keltjens et al. described the zinc-mediated propargylation reaction of 3-formylcephalosporins under aqueous Barbier-type



Scheme 63. Propargylation reaction of aldehydes



Scheme 64. Synthesis of highly selective propargylic and allenic alcohols



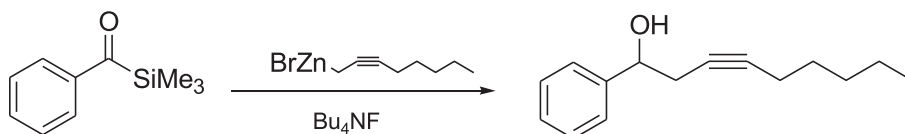
Scheme 65. Zinc-mediated propargylation reactions of aldimines

conditions, to produce the corresponding homopropargyl alcohols in good yields and with good-to-excellent diastereoselectivity. The aldehydes were reacted with propargyl bromides using nonactivated zinc in saturated aqueous NH_4Cl in THF at room temperature. In all cases the corresponding carbinols were formed in satisfactory yields and with high diastereoselectivities (Scheme 70) [95].

In 2014 Li-Juan Liu and Jin-Tao Liu reported for the first time the $Ti(Oi-Pr)_4$ and $CuCl$ catalyzed diastereoselective propargylation

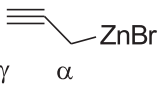
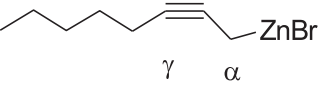
reaction of 2-chlorotetrafluoroethanesulfinimines with propargylzinc [126]. Zinc promoted asymmetric Barbier-type propargylation of aldimines was established using 2-Chlorotetrafluoroethane sulfonamide as the chiral auxiliary under mild conditions to give the corresponding homopropargylamines in good yields with up to 98% diastereoselectivity (Scheme 71).

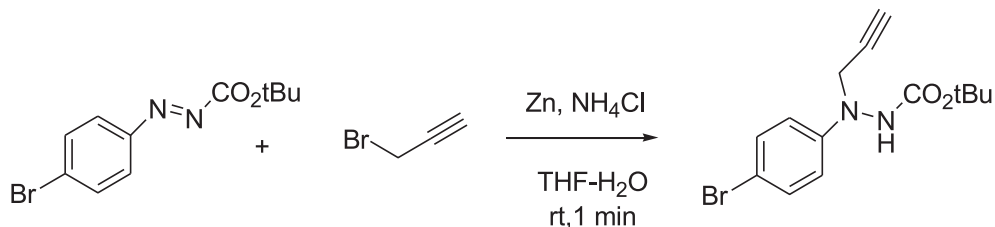
In very recent time Kobayashi and co-workers reported a zinc-amide catalyzed highly regioselective propargylation and allenylation reaction of various carbonyl compounds with allenyl boronate using $Zn(HMDS)_2$ ($HMDS$ = hexamethyldisilylamine) as a catalyst for the synthesis of a variety of tertiary allenyl or homopropargylic alcohols by simply varying the reaction conditions. The reaction proceeded at small amount of catalyst loading, and the tertiary homopropargyl and allenylic alcohols were selectively obtained in moderate to high yields, with broad substrate scope. High reactivity of the zinc amide species is the key for the smooth transmetalation and isomerization. Mechanistic studies suggest



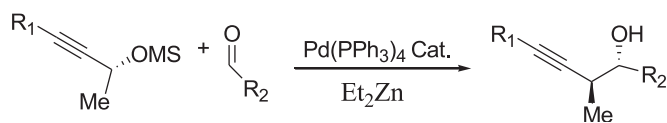
Scheme 66. Regioselective propargylation using acylsilane

Table 3
Regioselectivity of propargylation with aromatic and aliphatic aldehydes.

Nucleophiles	PhCHO α/γ	PhCOSiMe ₃ α/γ	PhCOSi(iPr) ₃ α/γ	Reactnt n-C ₅ H ₁₁ CHO α/γ	n-C ₅ H ₁₁ CO SiMe ₃ α/γ	n-C ₅ H ₁₁ COSi(iPr) ₃ α/γ
	92:8	93:7	>99:1	—	—	—
	35:65	>99:1	—	21:79	>99:1	—

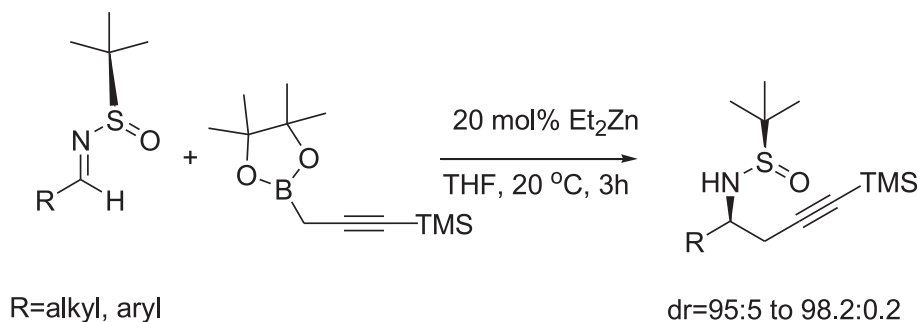


Scheme 67. Propargylation of tert-butyl 2-(4-bromophenyl)azocarboxylate ester

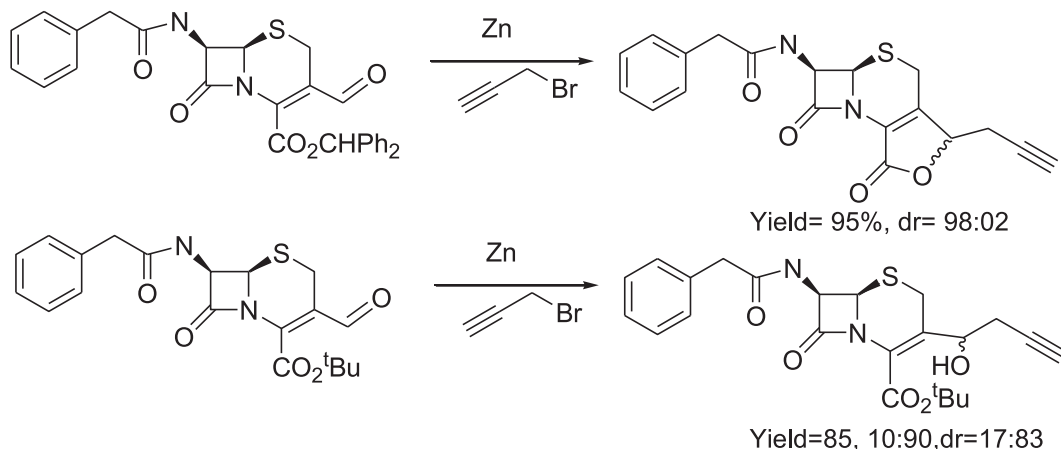


Scheme 68. Marshall type propargylations

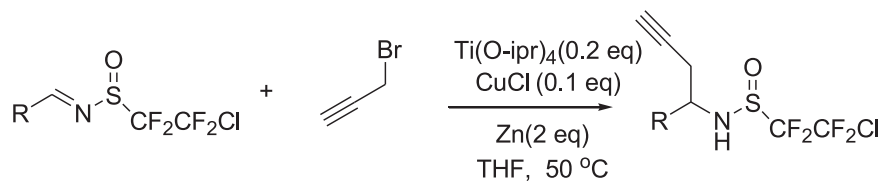
that transmetalation of allenyl boronate with zinc amide takes place to form a propargylzinc species under kinetic conditions, whereas the allenylzinc species results from isomerization of the propargylic species under thermodynamic conditions [127]. Mechanistic studies established that the reactions were controlled under kinetic and thermodynamic conditions (Scheme 72).



Scheme 69. Diastereoselective propargylation of tert-butanefulfinyl imines

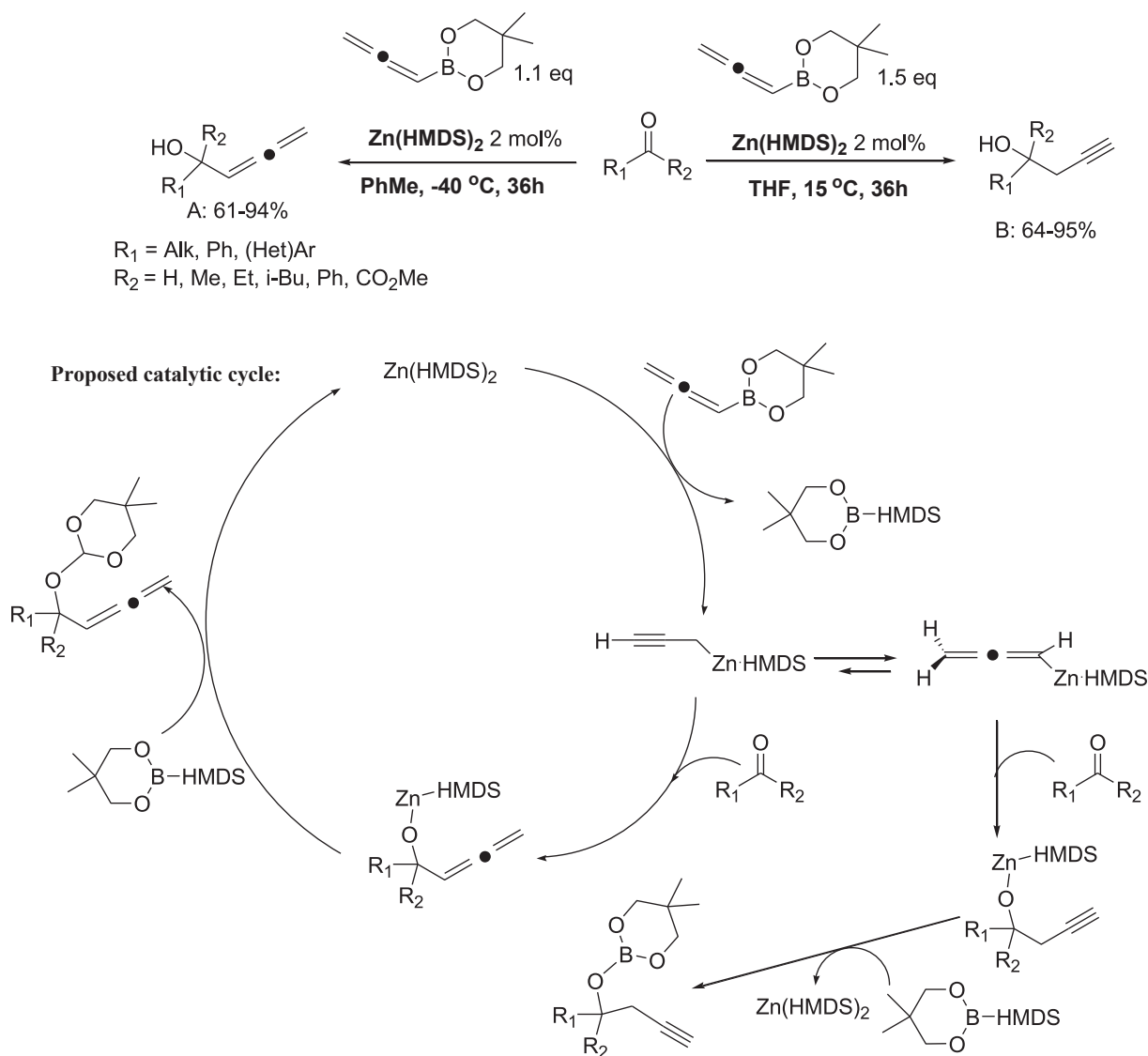


Scheme 70. Propargylation reactions of 3-formylcephalosporins

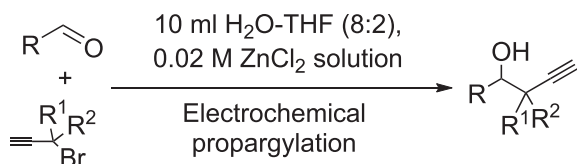


R= p-ClC₆H₄, 8h, (Yield=80%) de-80%
 p-ClC₆H₄, 8h, (Yield=90%) de-80%
 C₆H₅, 10h, (Yield=73%) de-84%
 m-ClC₆H₄, 8h, (Yield=66%) de-76%
 o-ClC₆H₄, 10h, (Yield=67%) de-92%
 p-FC₆H₄, 9h, (Yield=81%) de->96%
 o-FC₆H₄, 12h, (Yield=58%) de->96%
 p-BrC₆H₄, 14h, (Yield=61%) de-88%
 p-MeC₆H₄, 10h, (Yield=82%) de->98%

Scheme 71. Ti(Oi-Pr)₄ and CuCl catalyzed diastereoselective propargylation reactions of 2-chlorotetrafluoroethanesulfinimines



Scheme 72. Zinc-amide catalyzed highly regioselective propargylation and allenylation reactions



Scheme 73. Electrochemical synthesis of nano-zinc wire and its Reactivity in propargylation of aldehydes

Roy and co-workers have developed a new aqueous electrochemical process for propargylation of carbonyls using zinc chloride as the reagent (Fig. 4). Electrochemistry is used for propargylation of carbonyls in aqueous ZnCl_2 medium. For electrochemical process a unit galvanic cell was designed. ZnCl_2 was used as stoichiometric reagent and causes electrochemical deposition of zinc in cathode. Wire shaped nano zinc architecture has been formed in cathode during electrochemical process which is the active reagent. Homopropargylic alcohols are synthesized in good yields. After the organic reaction is over aqueous solution containing zinc salts can be reused up to 5th cycle without significant loss of reactivity (Scheme 73) [128].

4. Conclusion

The development and use of organozinc reagents in synthesis has been an underlying theme of modern organic synthesis. It is evident from the above account that efficient methodologies now exist for the preparation of organozinc reagents. However, there are still needs to develop simple methods for their preparation, for example using zinc dust in THF or water as solvent. Moreover, less reactive alkyl chlorides and aryl bromides are still useless substrates for the reaction with zinc.

Organozincs are reactive species and have proved their utility in asymmetric synthesis. However there are very few catalytic protocols for their use in enantioselective transformations. To sum up, the oldest organometallic reagent still remains significantly unexplored, and promises rich dividend for researchers.

Declaration of competing interest

The authors do not have any conflict of interest.

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